US012404924B2

(12) **United States Patent**
Mason et al.

(10) **Patent No.: US 12,404,924 B2**
(45) **Date of Patent: Sep. 2, 2025**

(54) **DEVICES AND METHOD FOR REGULATING COOLER FLOW THROUGH AUTOMOTIVE TRANSMISSIONS**

(71) Applicant: **SUPERIOR TRANSMISSION PARTS, INC.**, Tallahassee, FL (US)

(72) Inventors: **Dean Mason**, Tallahassee, FL (US); **Victor Castano**, Tallahassee, FL (US)

(73) Assignee: **SUPERIOR TRANSMISSION PARTS, INC.**, Tallahassee, FL (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 244 days.

(21) Appl. No.: **18/220,497**

(22) Filed: **Jul. 11, 2023**

(65) **Prior Publication Data**

US 2023/0349460 A1 Nov. 2, 2023

Related U.S. Application Data

(63) Continuation-in-part of application No. 17/288,720, filed as application No. PCT/US2019/058831 on Oct. 30, 2019, now Pat. No. 11,725,891.

(60) Provisional application No. 63/458,465, filed on Apr. 11, 2023, provisional application No. 62/752,539, filed on Oct. 30, 2018.

(51) **Int. Cl.**
F16H 57/00 (2012.01)
F01P 7/14 (2006.01)
F16H 57/04 (2010.01)

(52) **U.S. Cl.**
CPC ***F16H 57/0413*** (2013.01); ***F01P 7/14*** (2013.01); *F01P 2007/146* (2013.01)

(58) **Field of Classification Search**
CPC F28F 27/02; F28F 2250/06; F28F 2265/12; F16K 15/026; F16K 15/028; F16K 15/03; F16K 17/04; F16K 17/0406; F16K 17/30; F16K 3/265; F16H 57/0413; F01P 7/14; F01P 2007/146
See application file for complete search history.

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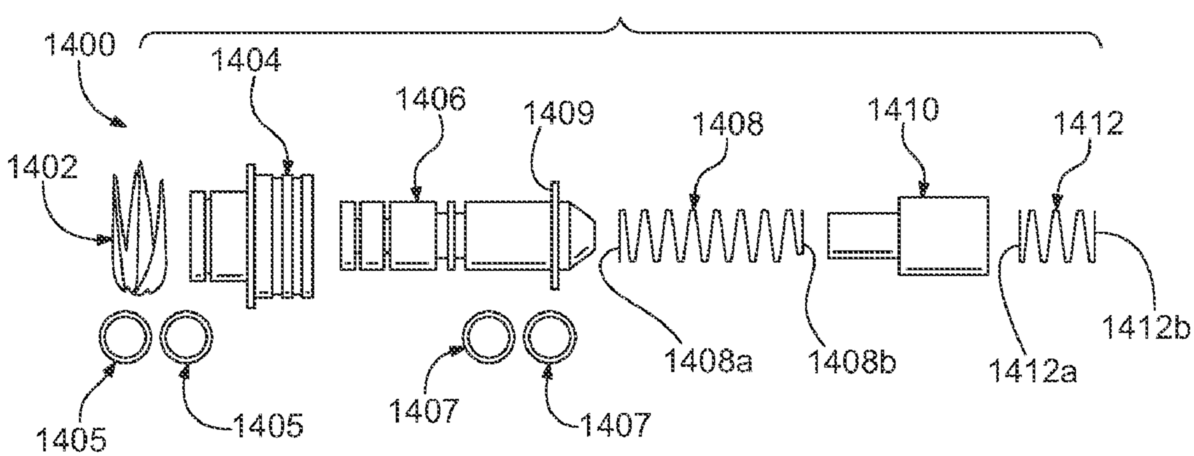
Primary Examiner — Travis Ruby

(57) **ABSTRACT**

Methods and valves for providing a continuous flow of cooler fluid in a fluid circuit of a thermal control system between a cooler and automotive transmission such that free flow of cooler fluid between the cooler and transmission exists at vehicle start-up. Fluid flow to and from the cooler is bypassed in case of pressure increases in cooler lines or pressure differentials, for example cause by a blockage in the cooler, such that the cooler fluid flow bypasses the cooler and continues in the fluid circuit through a thermal element of the thermal control system and back to the transmission.

13 Claims, 24 Drawing Sheets



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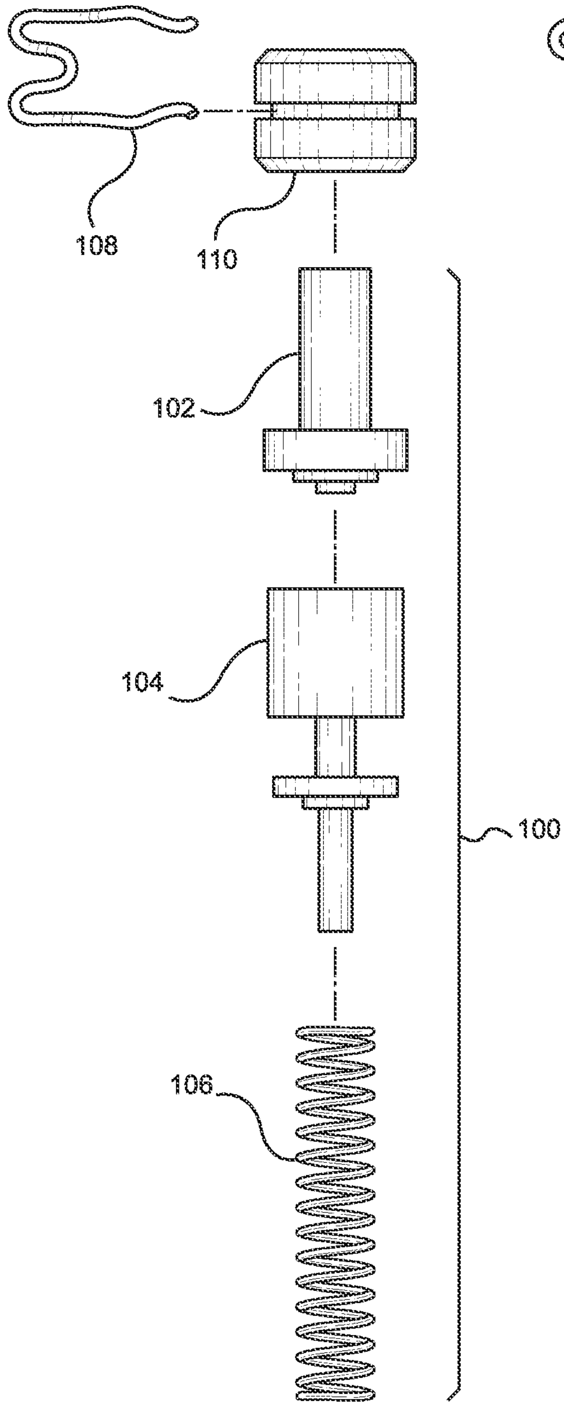


FIG. 1
Prior Art

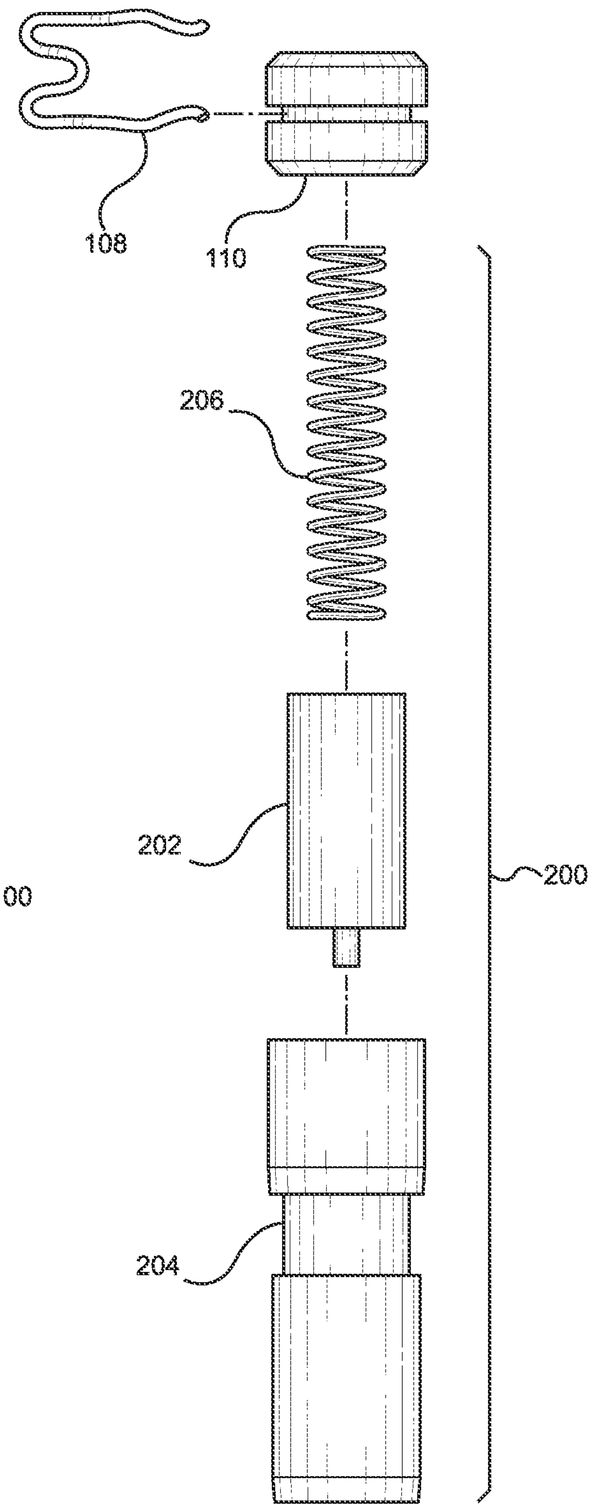


FIG. 2

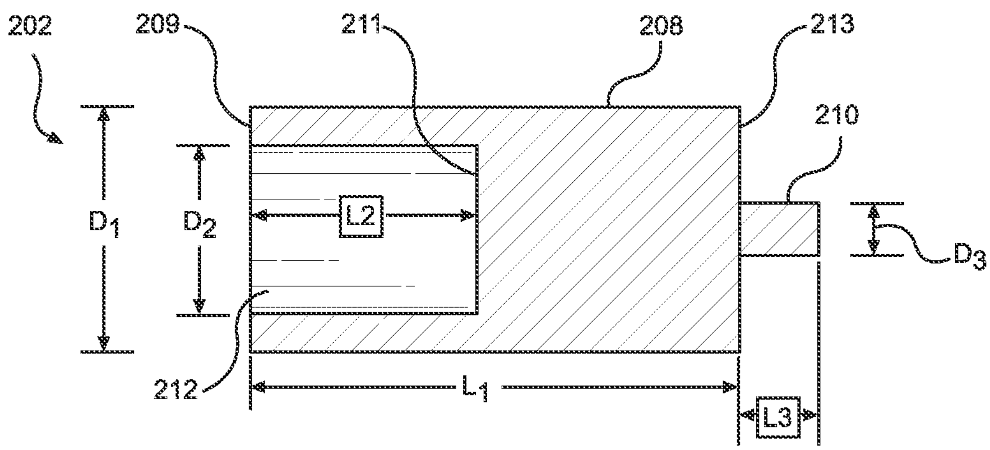

202
209
211
208
213
210
D1
D2
L2
D3
212
L1
L3

FIG. 3

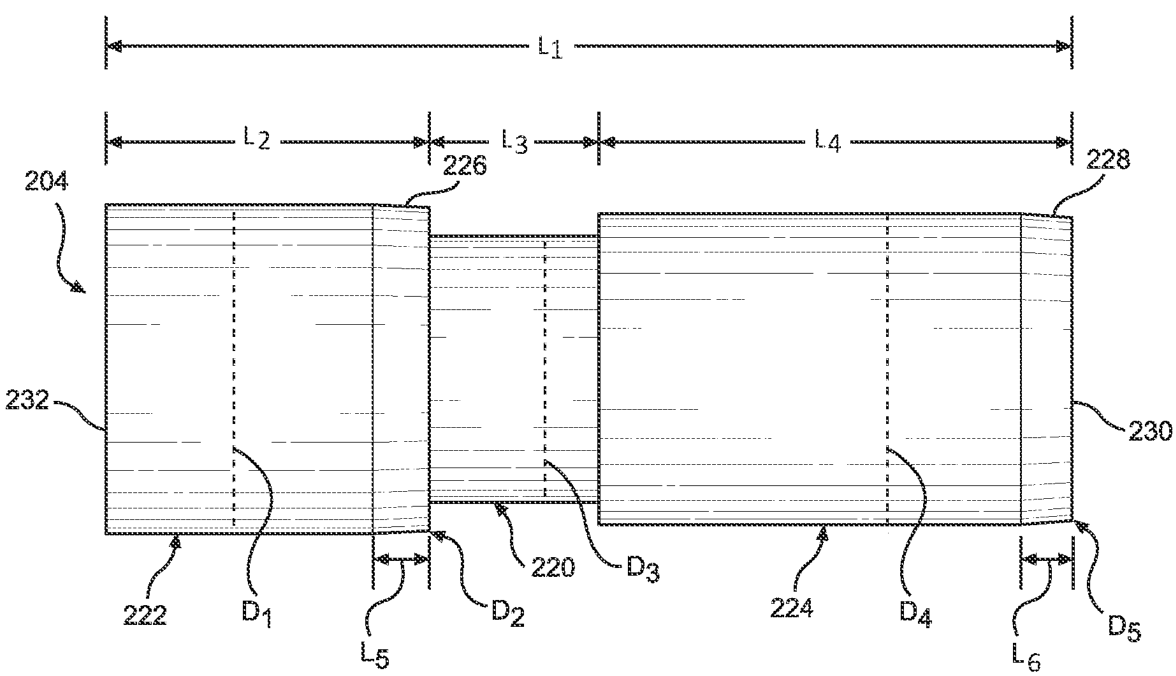

L1
L2
L3
L4
226
228
204
232
230
222
D1
L5
D2
220
D3
224
D4
L6
D5

FIG. 4

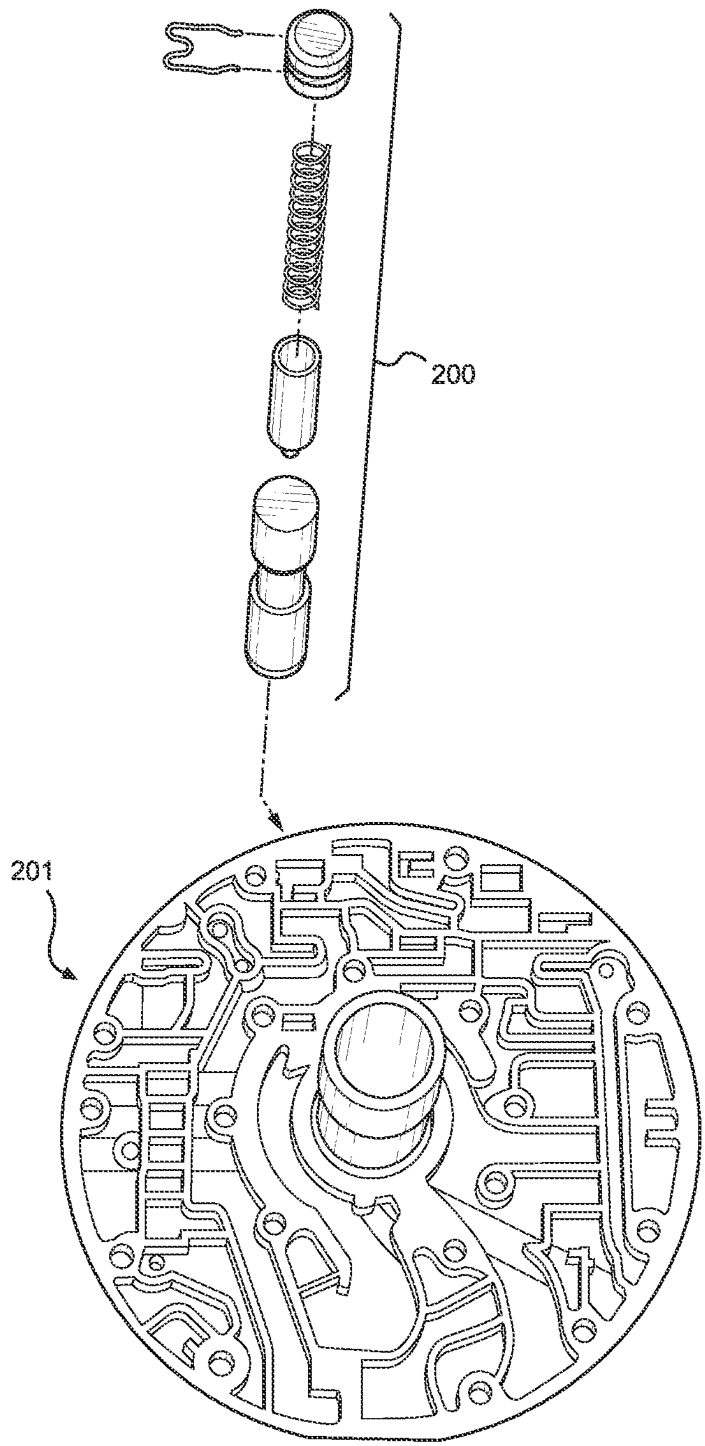


FIG. 5

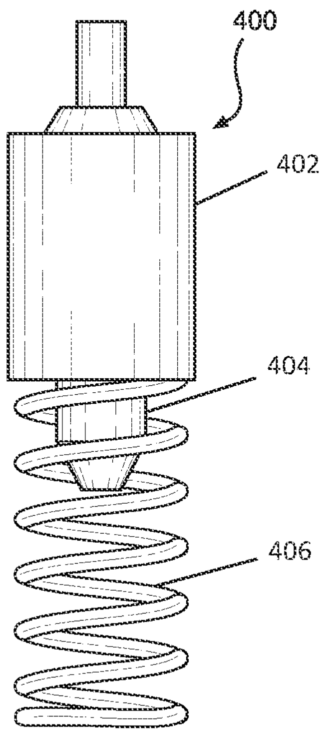


FIG. 6
Prior Art

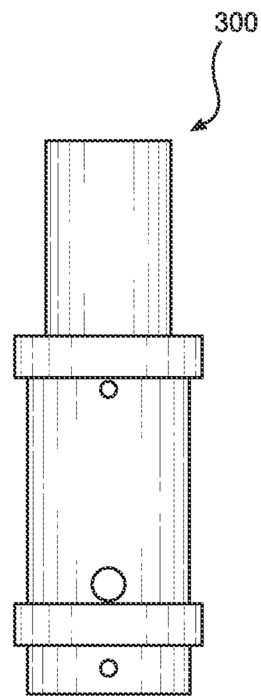


FIG. 7

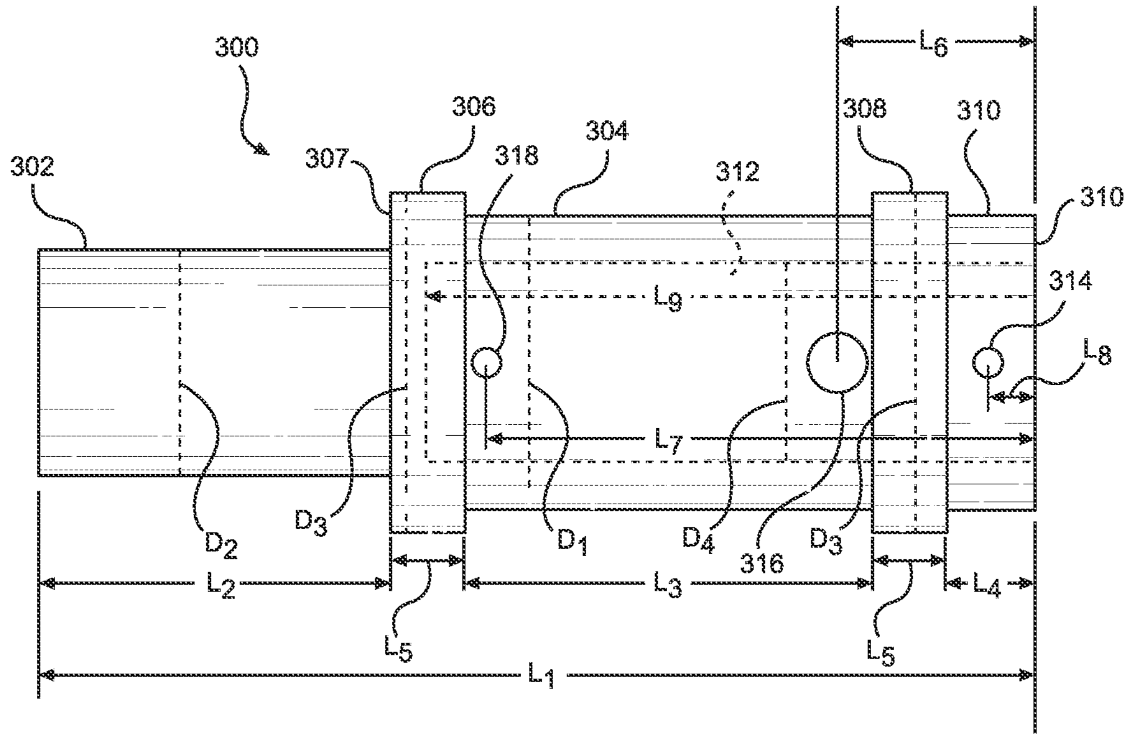

300
302
307
306
318
304
312
308
310
310
314
L_6
L_9
L_8
L_7
D_3
D_2
D_1
D_4
316
D_3
L_2
L_5
L_3
L_5
L_4
L_1

FIG. 8

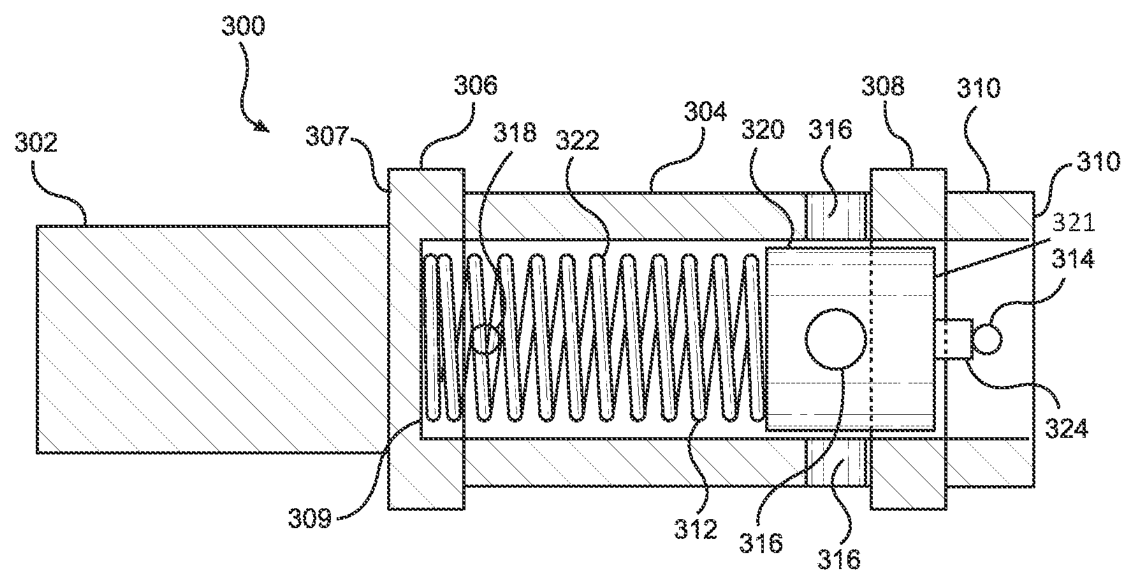

300
302
307
306
318
322
304
320
316
308
310
310
321
314
324
309
312
316
316

FIG. 9

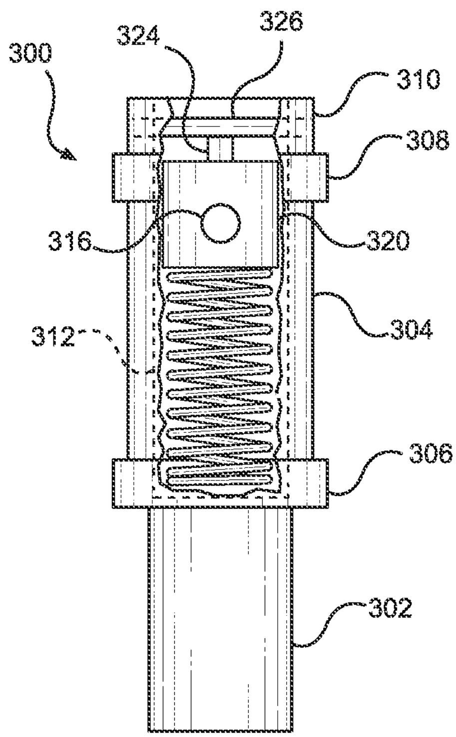
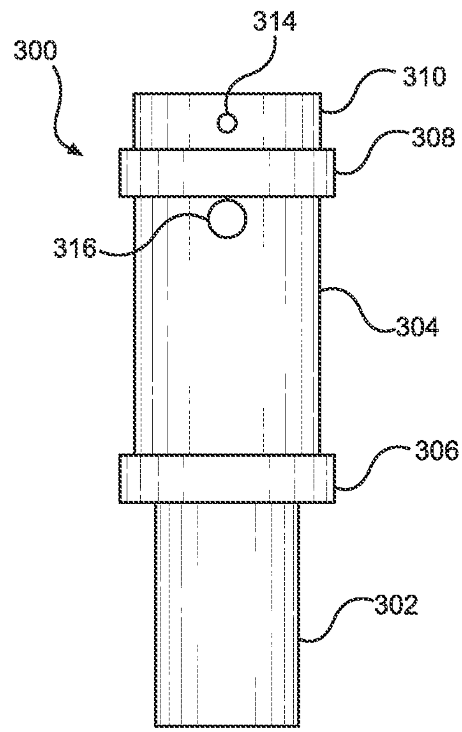


FIG. 10



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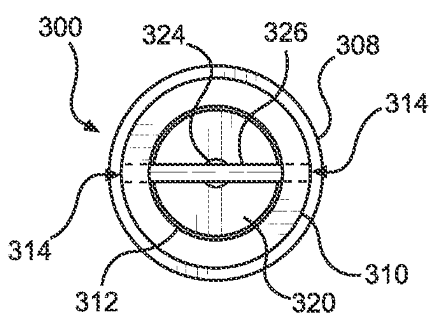


FIG. 12

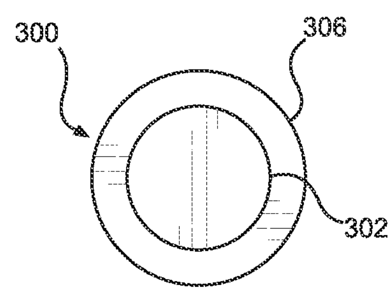


FIG. 13

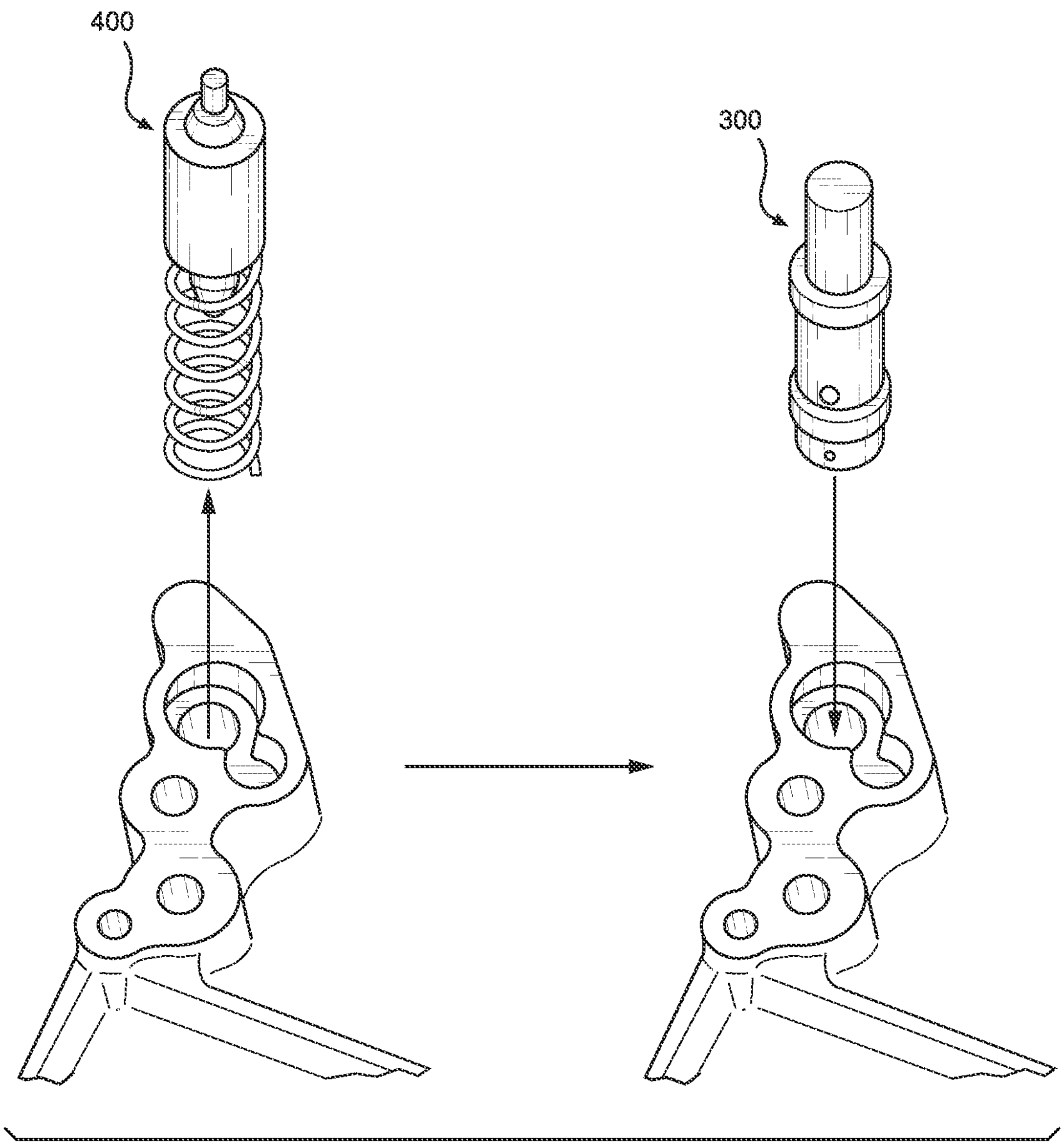


FIG. 14

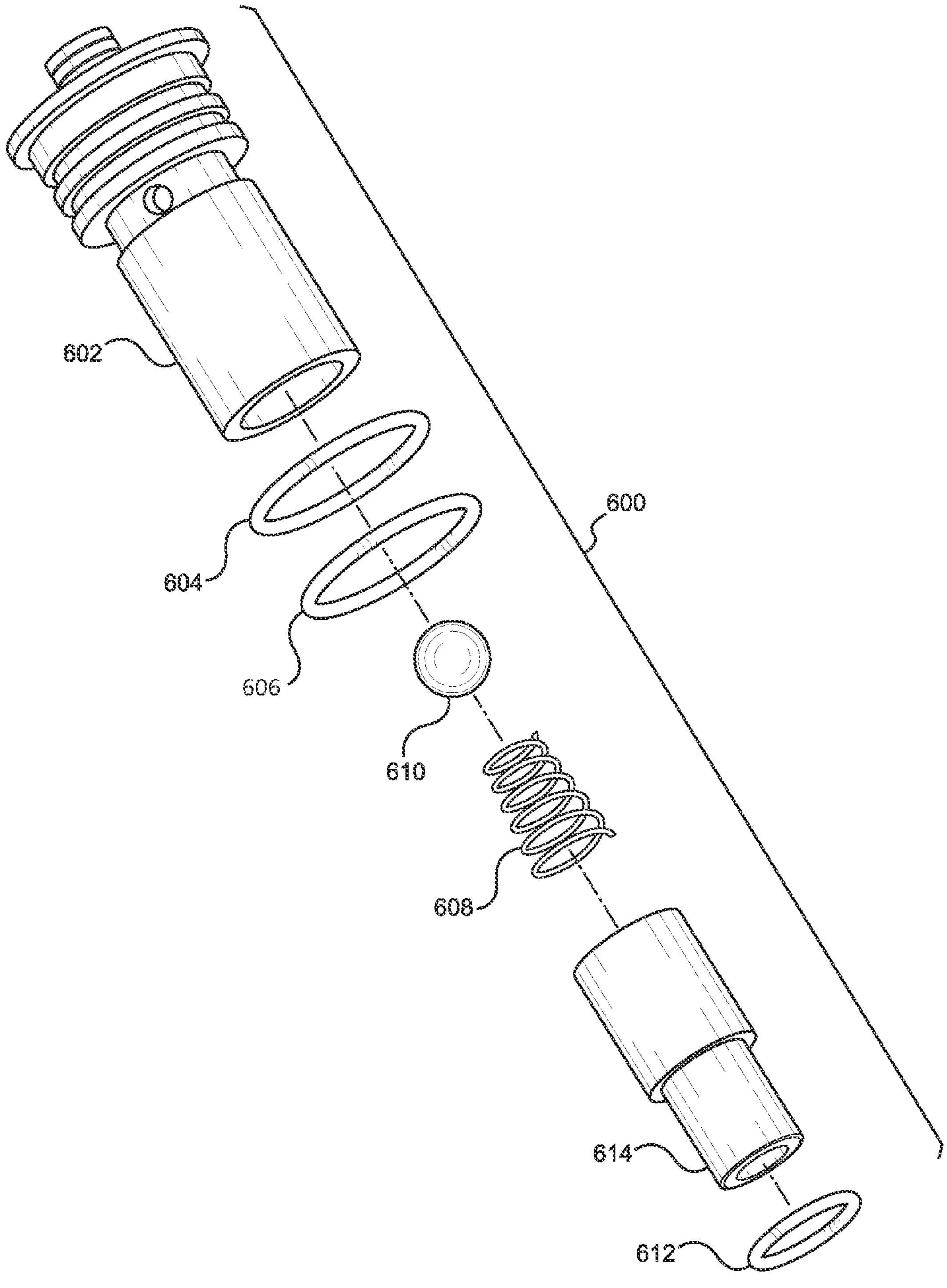


FIG. 15

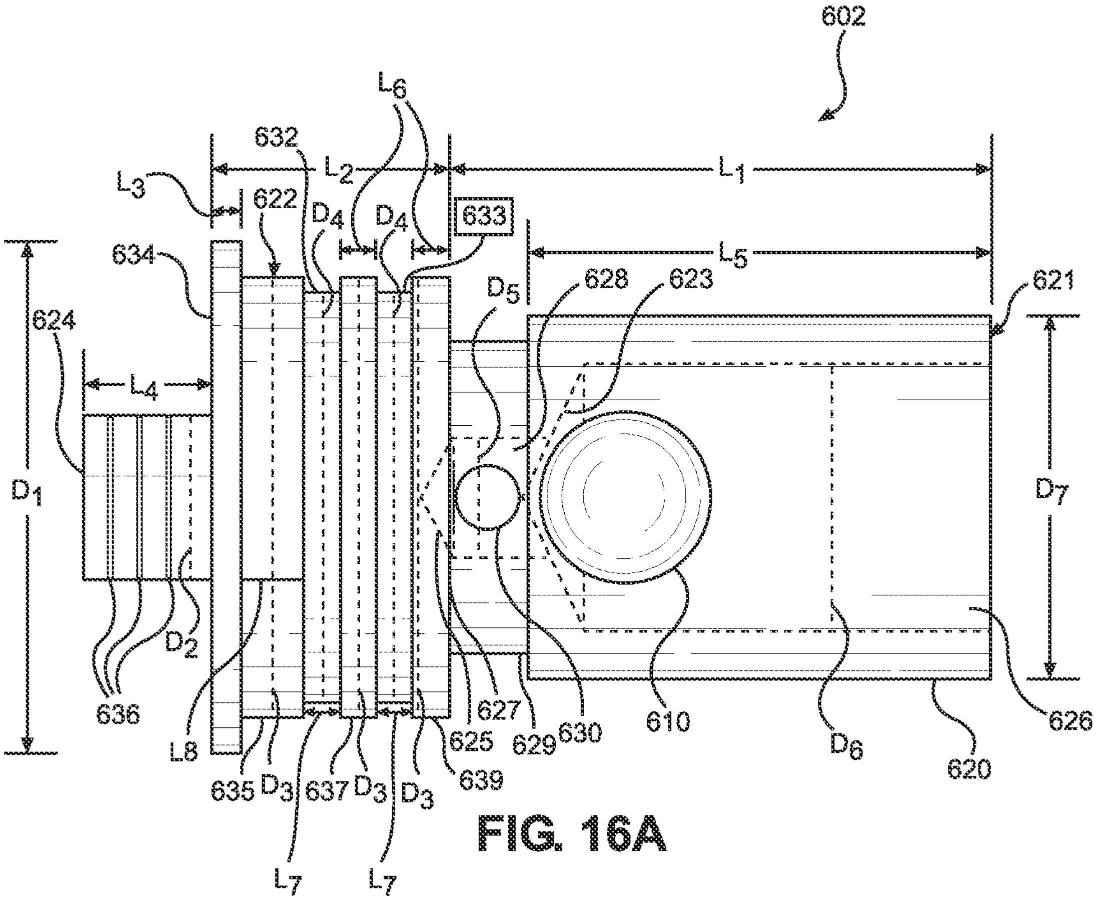


FIG. 16A

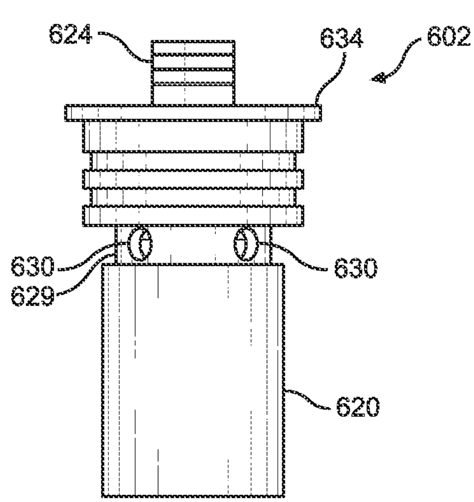


FIG. 16B

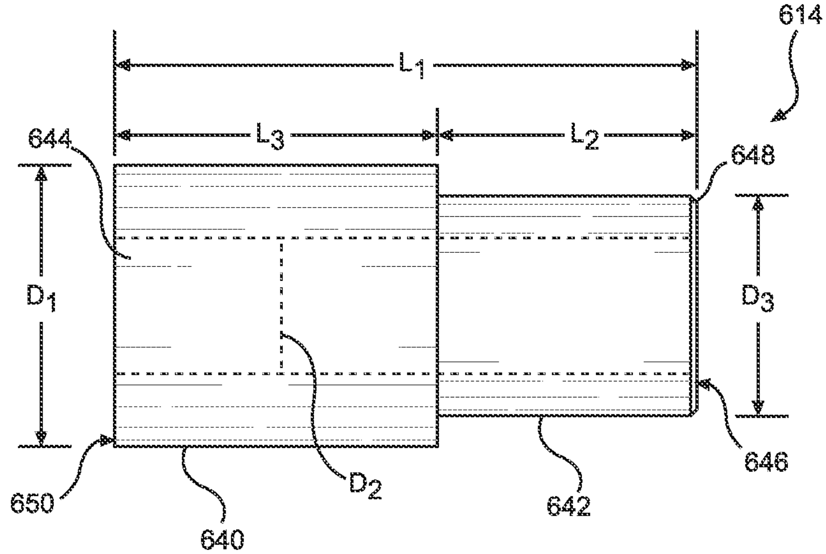
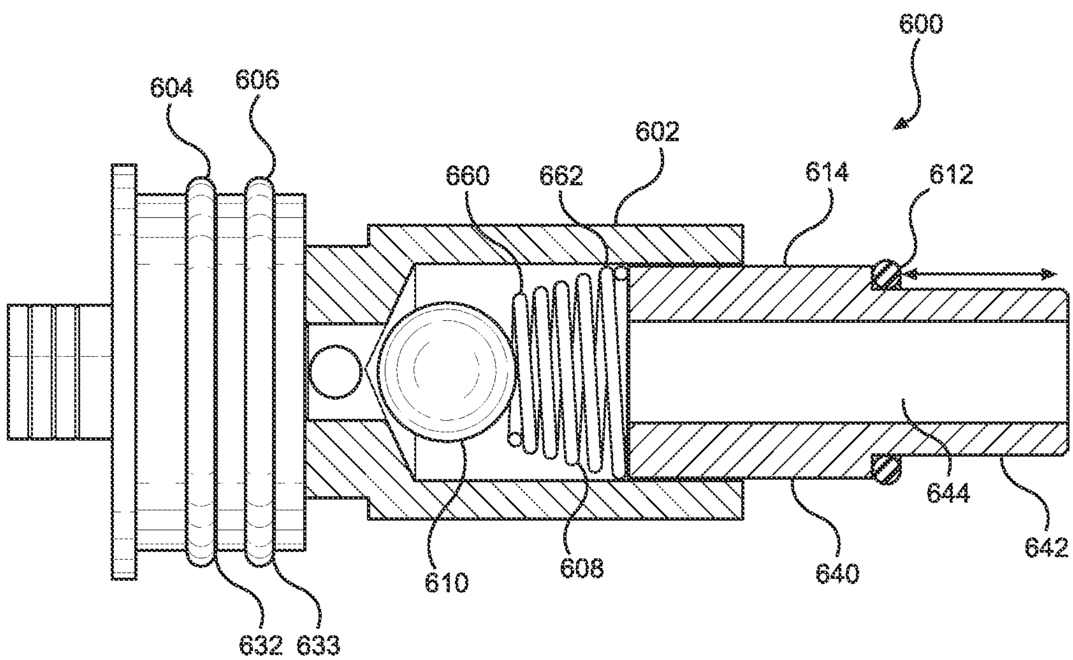

614
L1
L3
L2
644
648
D1
D3
D2
650
640
642
646

FIG. 17


600
604
606
602
660
662
614
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644
642
640
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610
632
633

FIG. 18

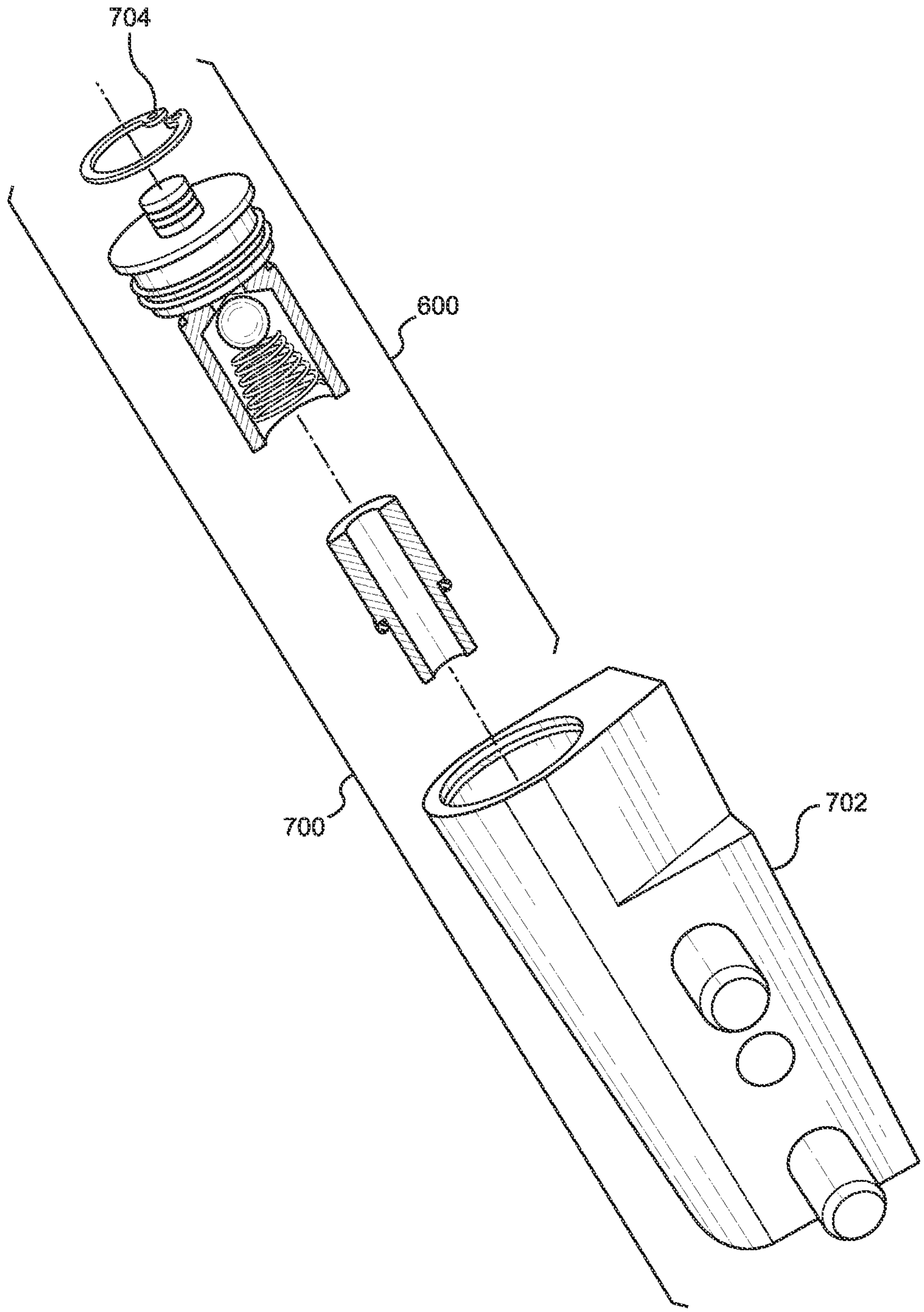


FIG. 19

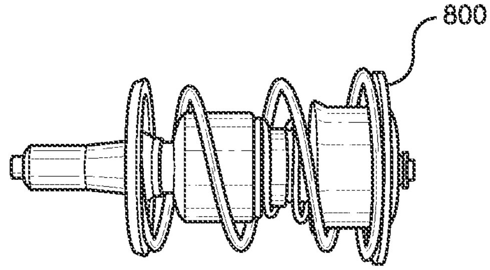


FIG. 20A
PRIOR ART

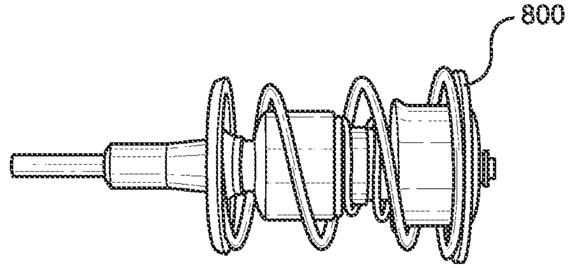


FIG. 21A
PRIOR ART

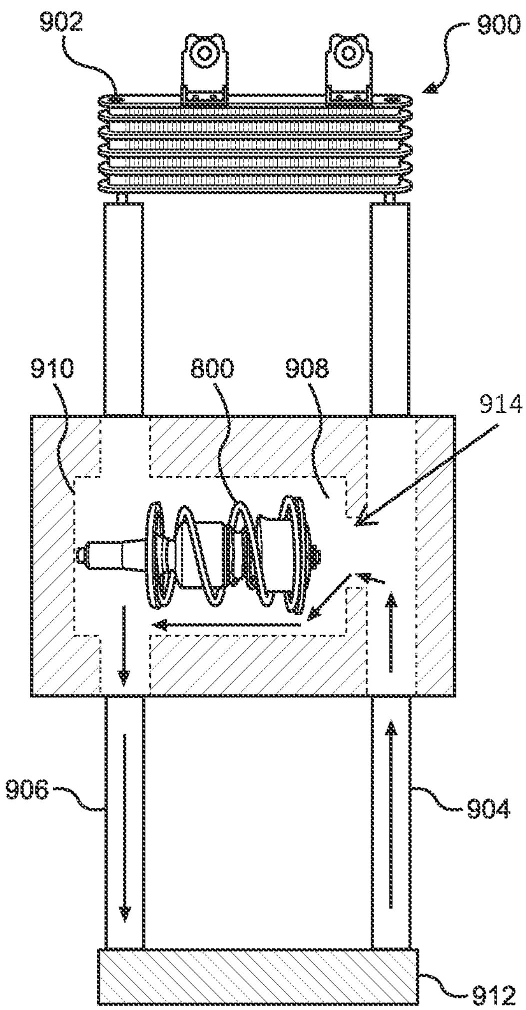


FIG. 20B
PRIOR ART

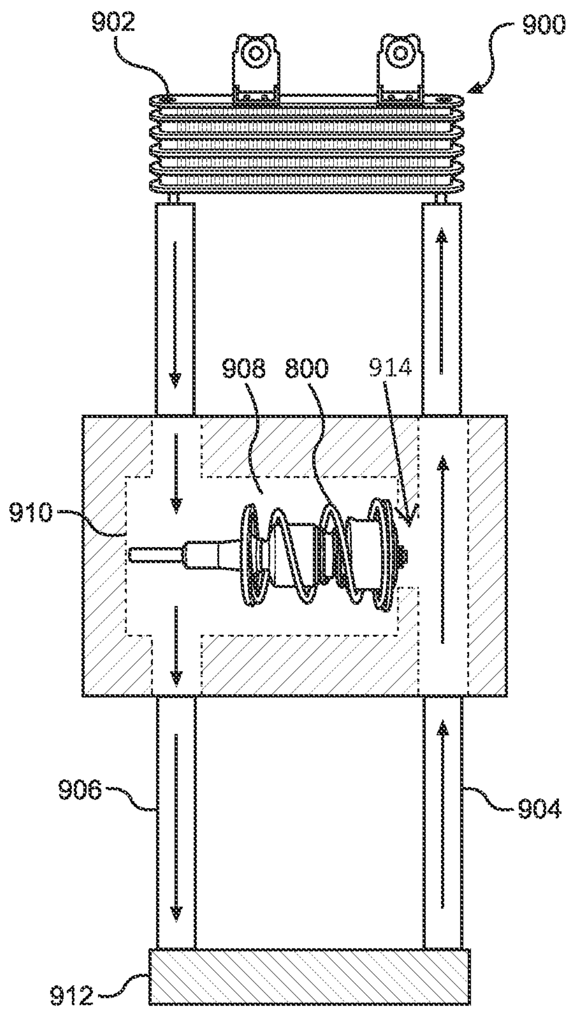
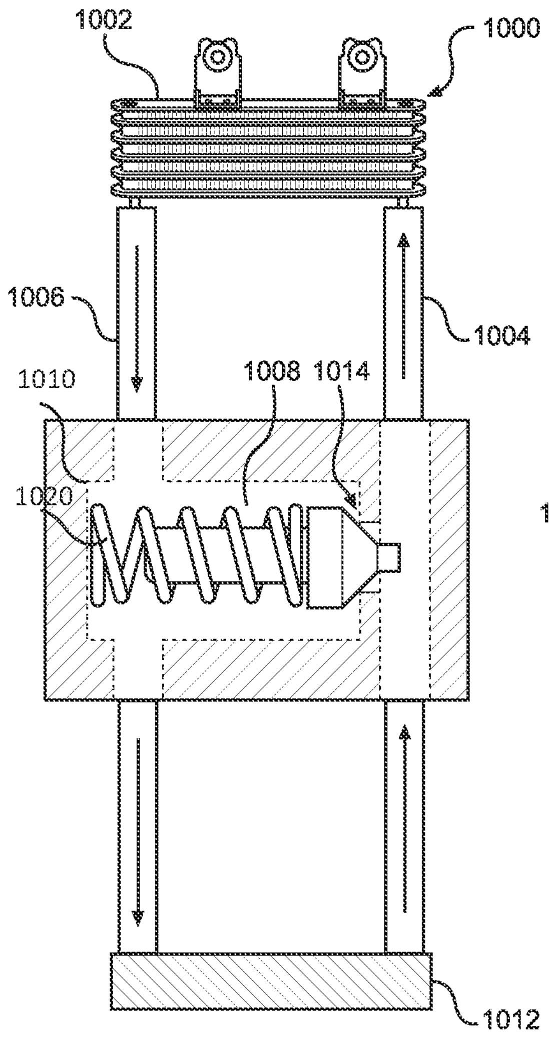
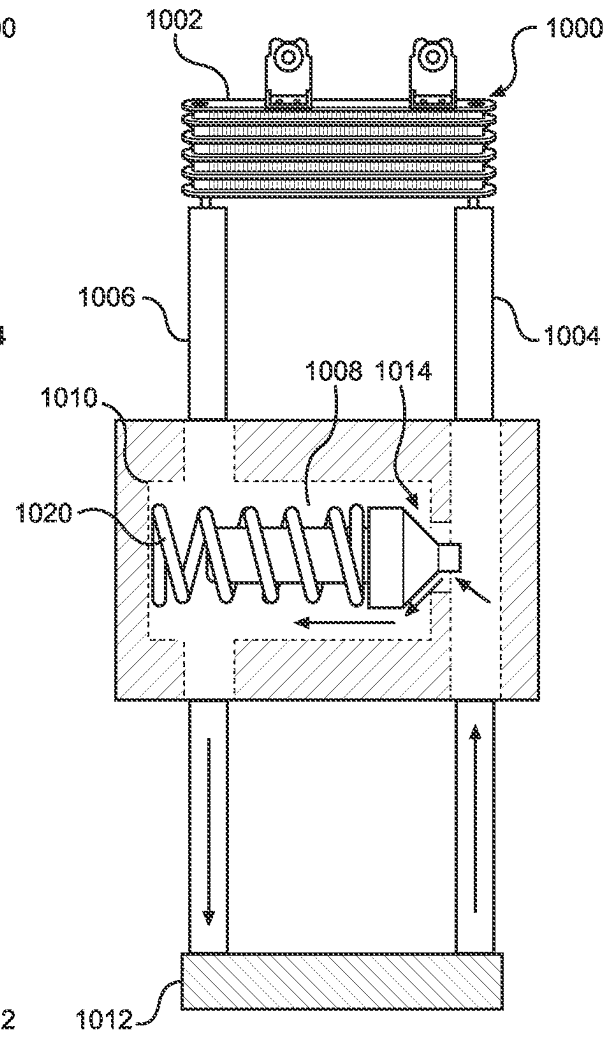


FIG. 21B
PRIOR ART



1002
1000
1006
1004
1010
1008
1014
1020
1012

FIG. 22



1002
1000
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1004
1010
1008
1014
1020
1012

FIG. 23

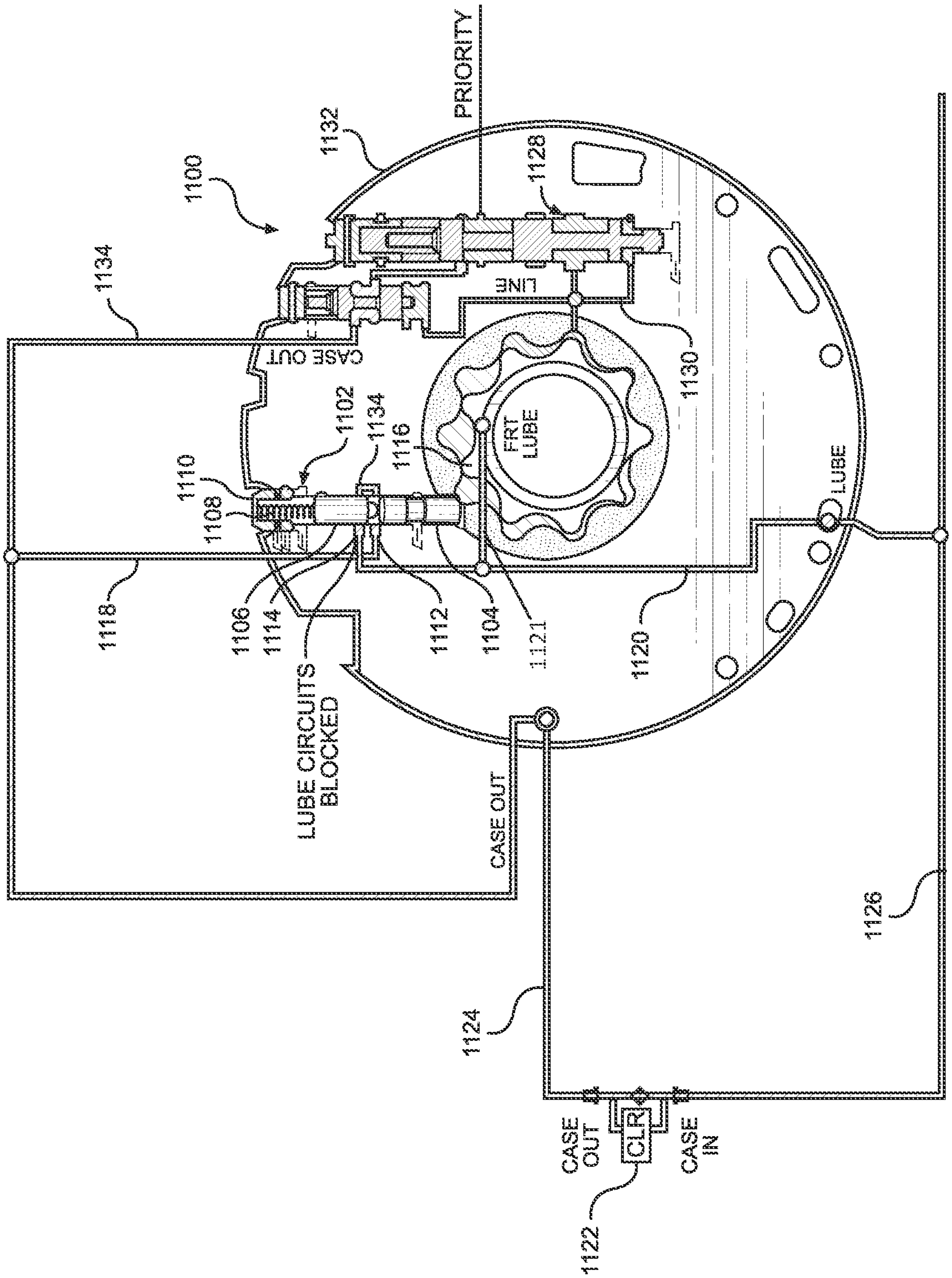
PRIORITY
1132
1128
1100
1134
LINE
CASE OUT
1130
FRT LUBE
1102
1134
1116
1110
1108
LUBE
1118
1106
1114
1112
1104
1121
1120
LUBE CIRCUITS BLOCKED
CASE OUT
1126
1124
CASE OUT
CLR
CASE IN
1122

FIG. 24

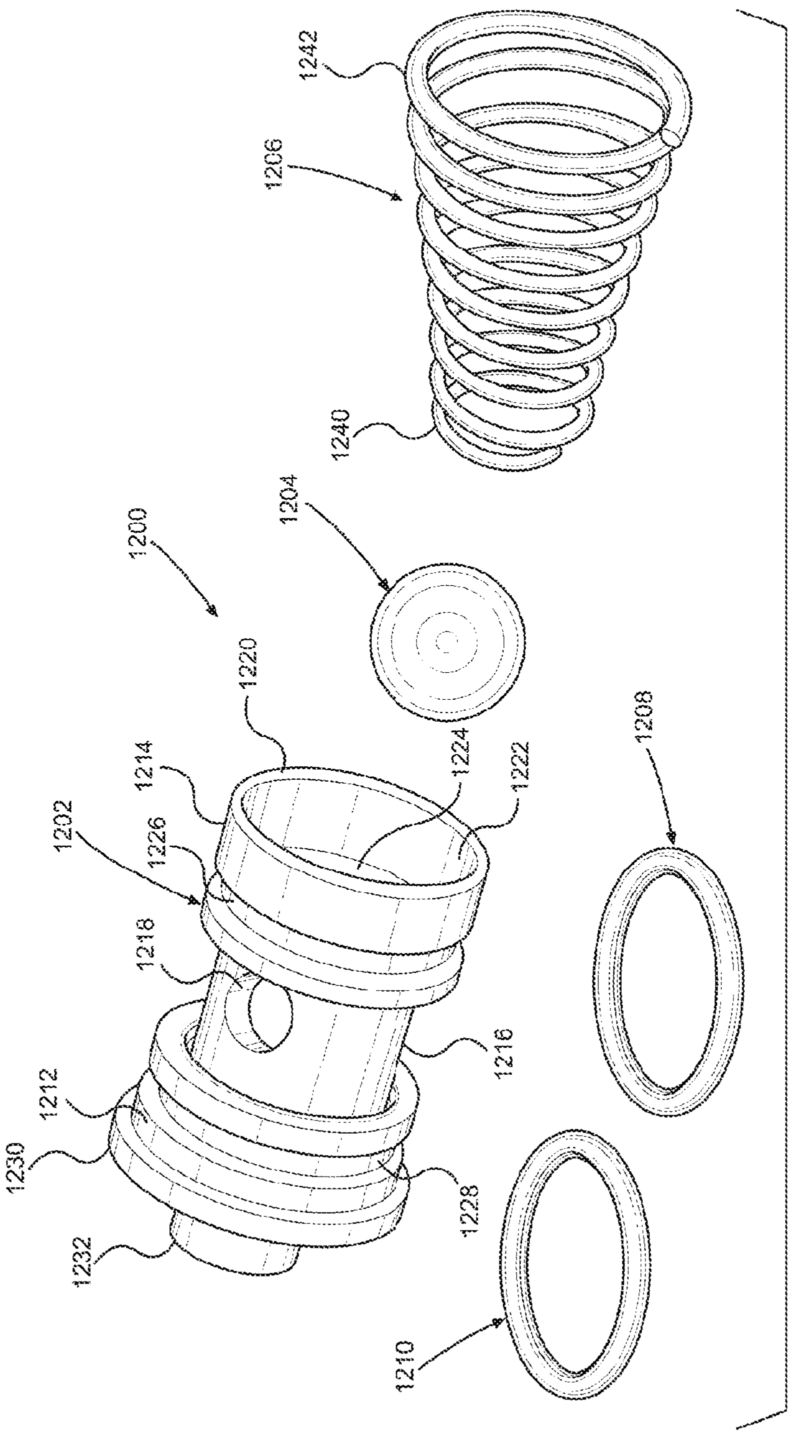
1242
1206
1240
1204
1200
1220
1214
1224
1222
1202
1226
1218
1216
1212
1230
1228
1232
1210
1208

FIG. 25

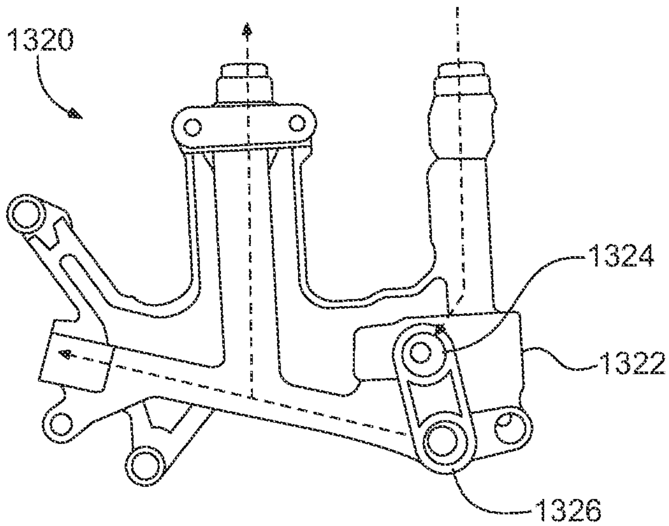

FIG. 26A

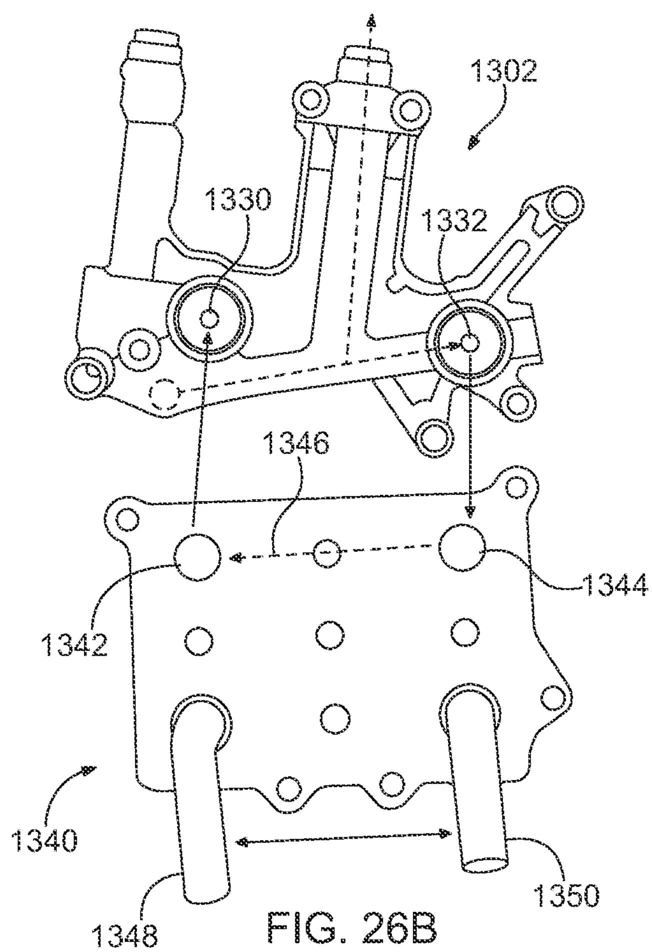

FIG. 26B

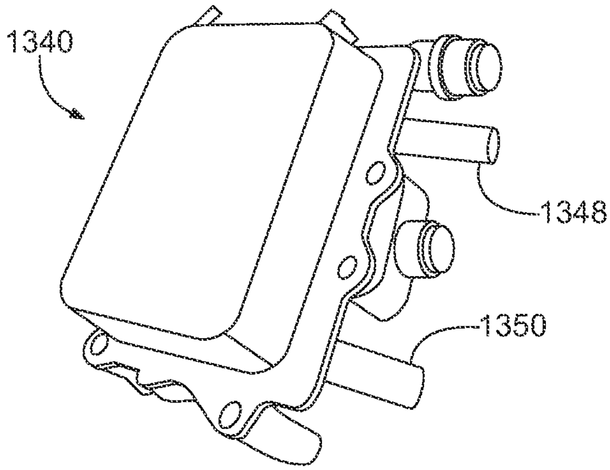

FIG. 26C

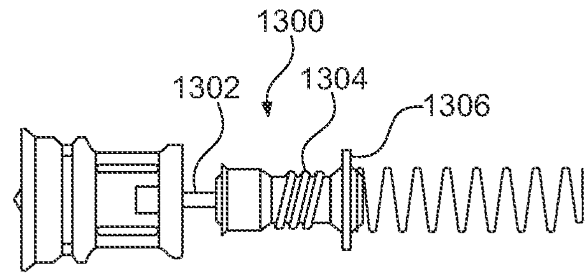

FIG. 26D
(PRIOR ART)

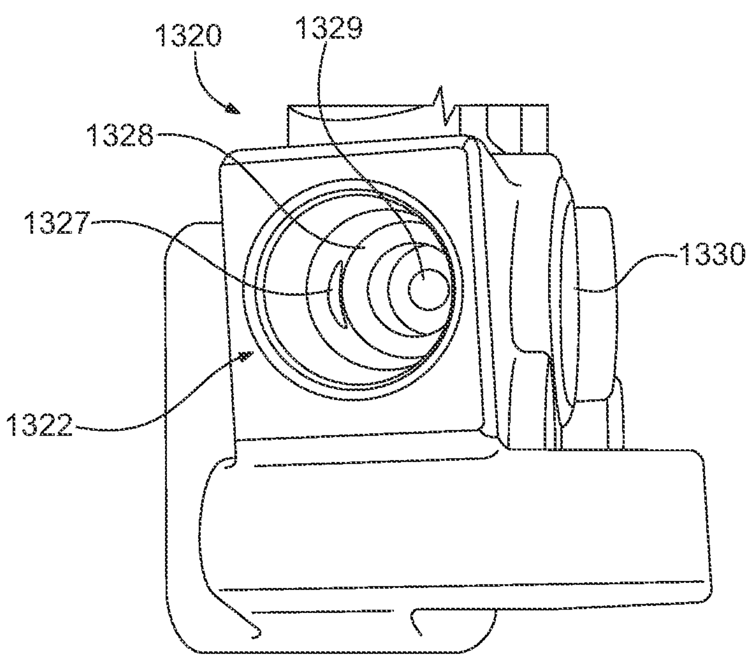

FIG. 26E

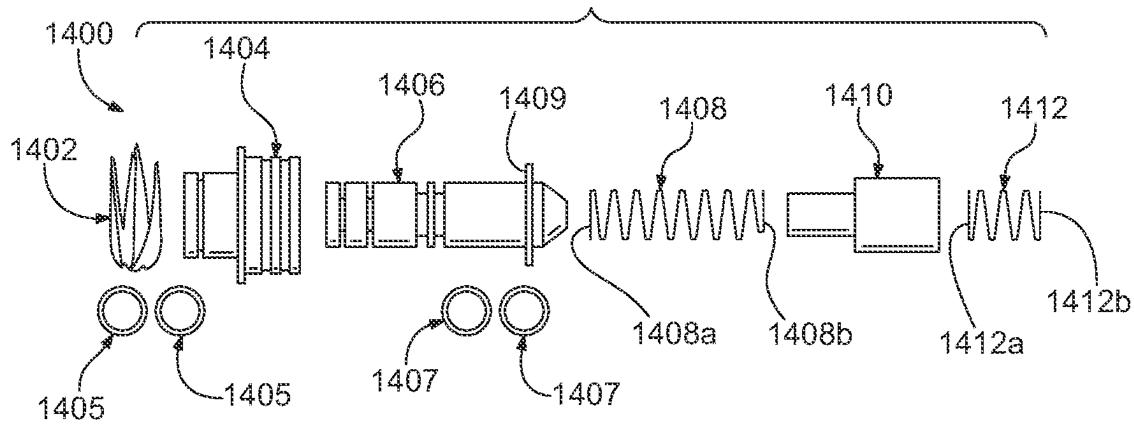
1400
1404
1406
1409
1408
1410
1412
1402
1408a
1408b
1412a
1412b
1405
1405
1407
1407

FIG. 27

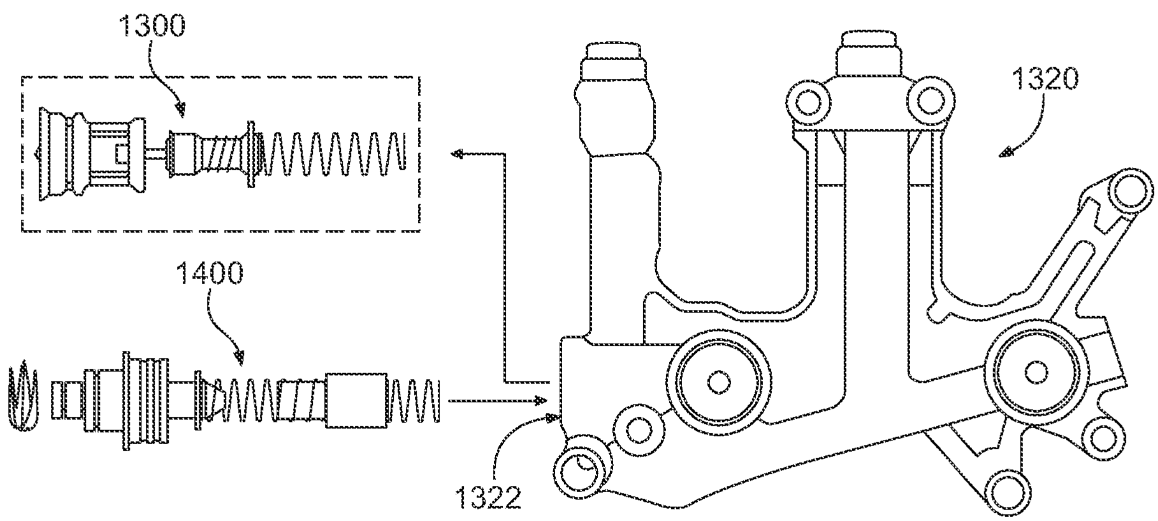
1300
1320
1400
1322

FIG. 28

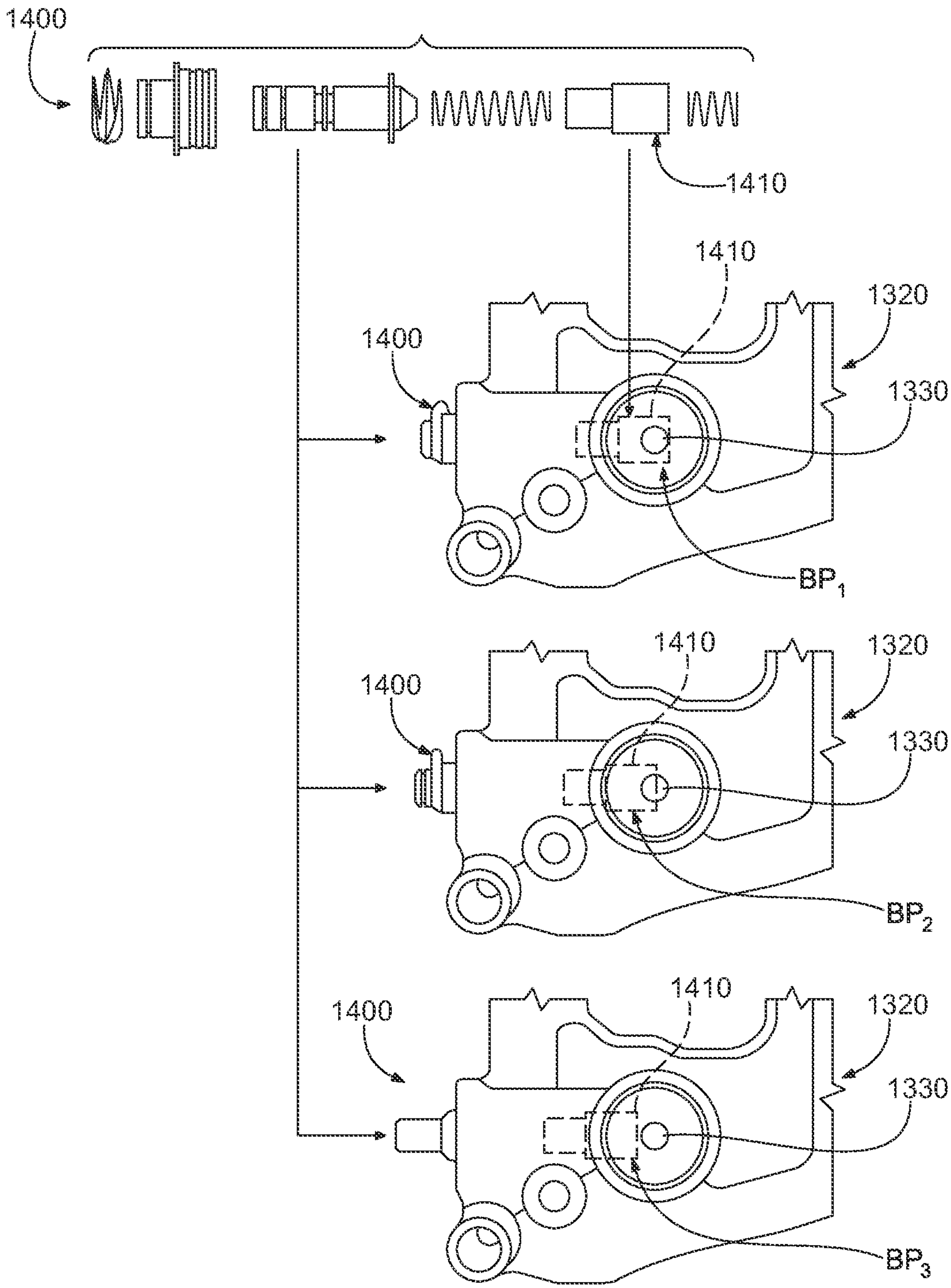
1400
1410
1410
1320
1400
1330
BP1
1410
1320
1400
1330
BP2
1410
1320
1400
1330
BP3

FIG. 29

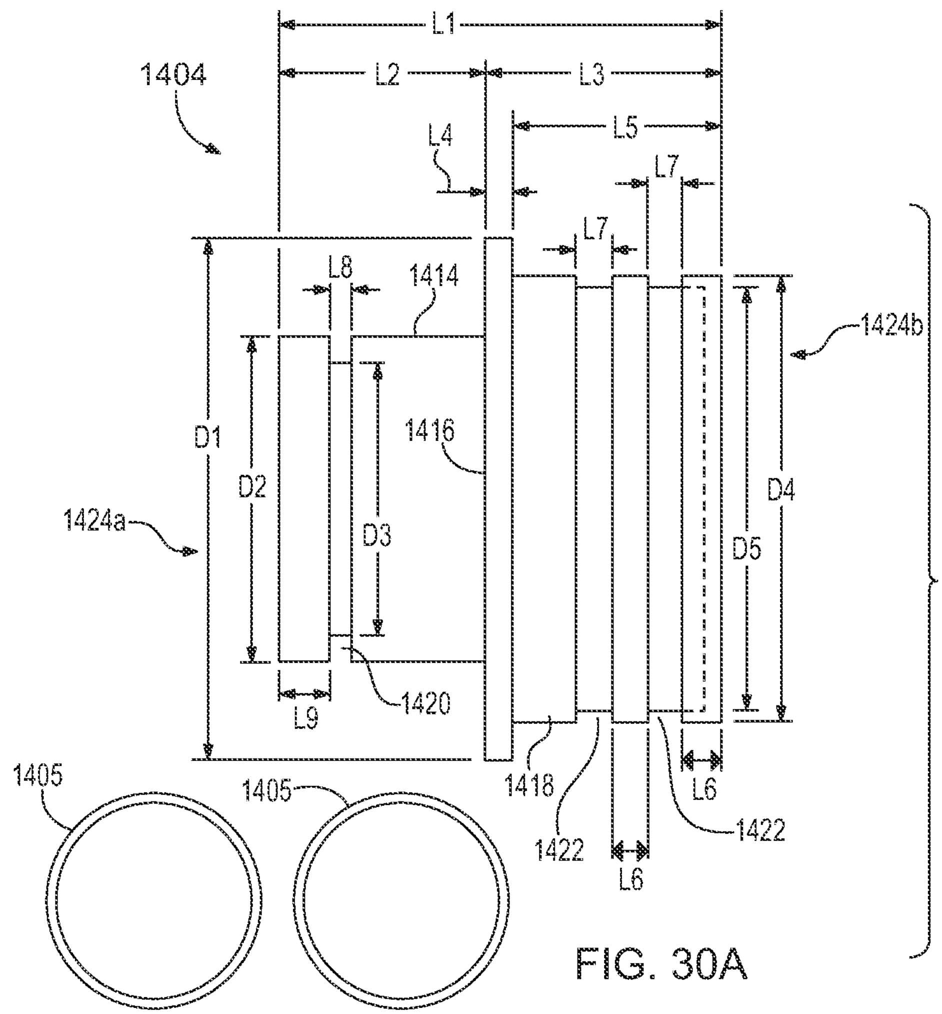

FIG. 30A

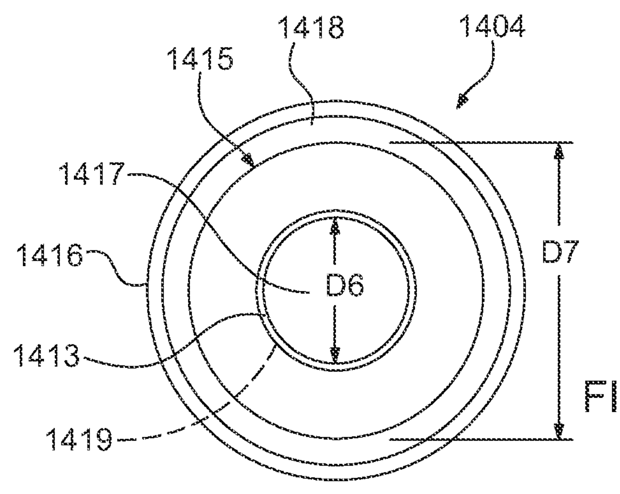

FIG. 30B

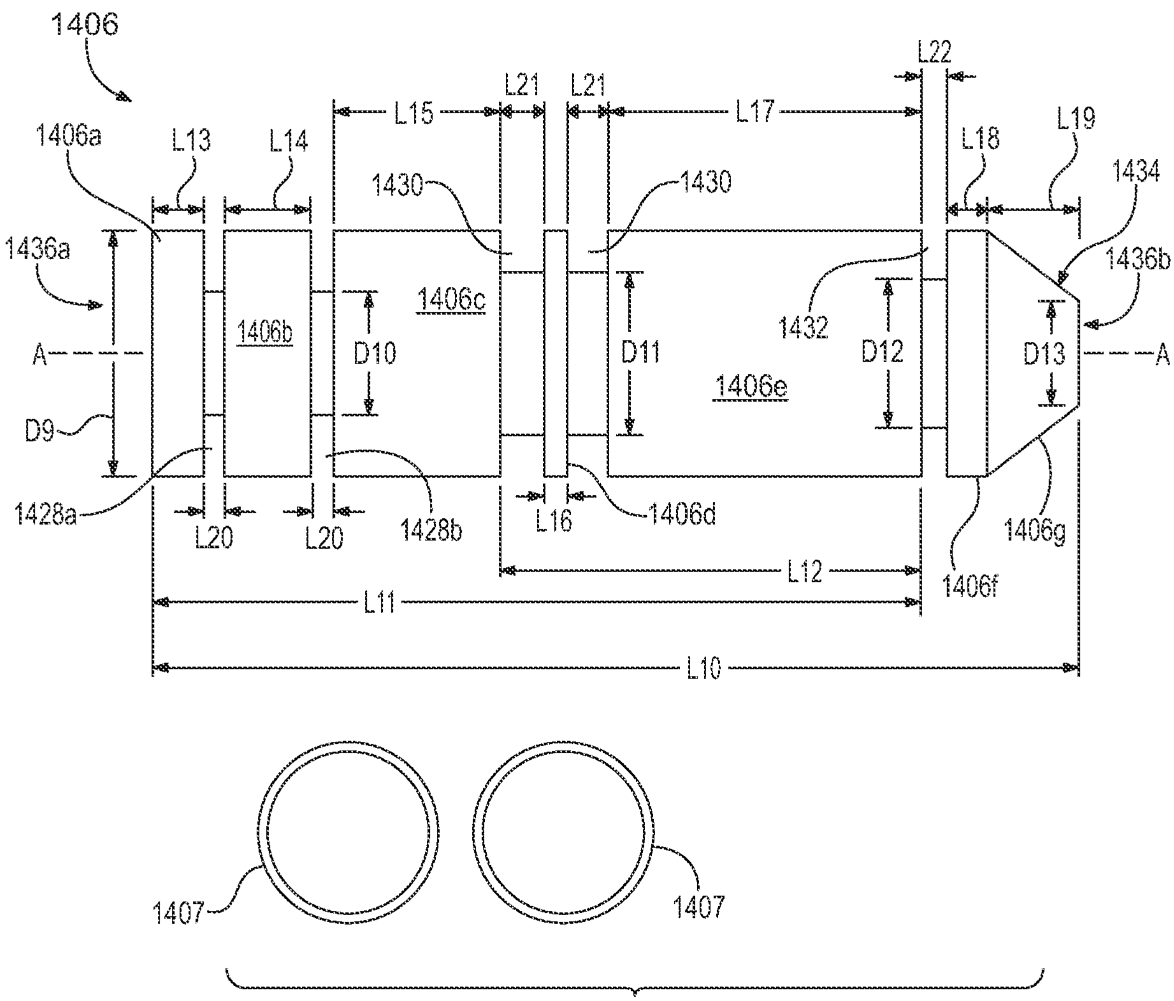
1406
1406a
L13
L14
L15
L21
L21
L17
L22
L18
L19
1434
1436a
1430
1430
1436b
1406b
1406c
D10
D11
1432
D12
D13
A
A
1406e
D9
1428a
L20
L20
1428b
L16
1406d
1406g
L12
1406f
L11
L10
1407
1407

FIG. 30C

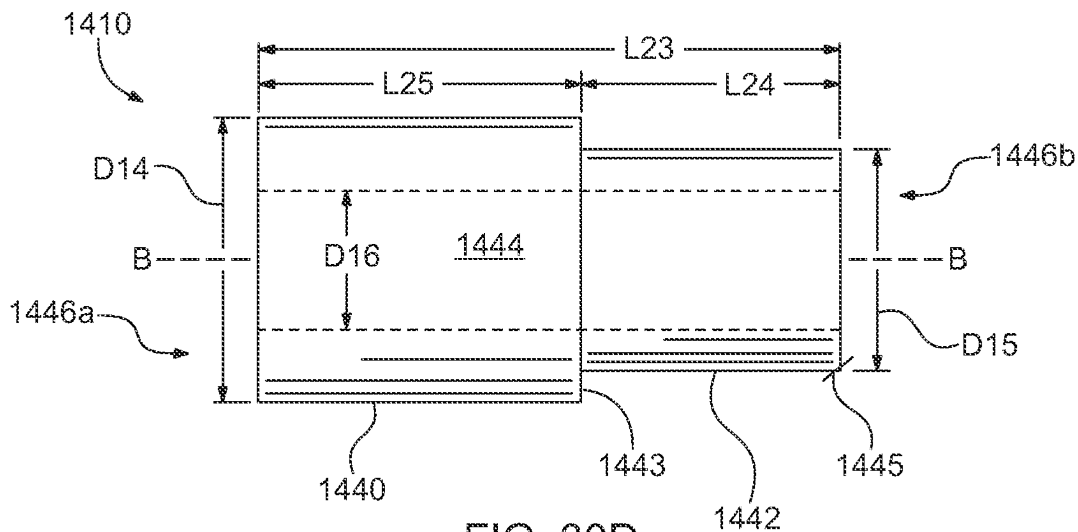

FIG. 30D

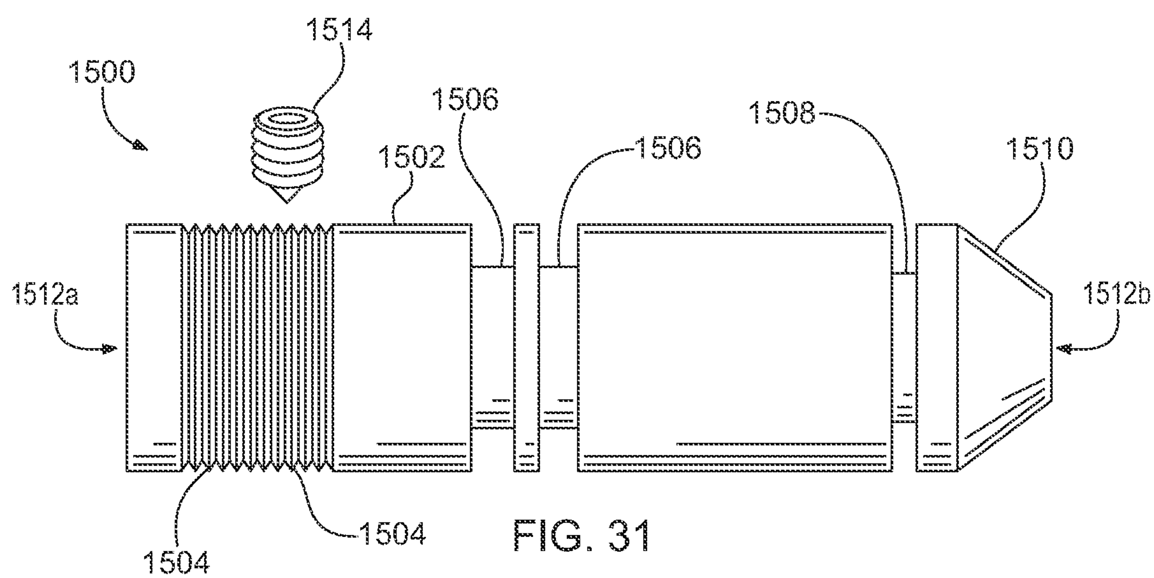

FIG. 31

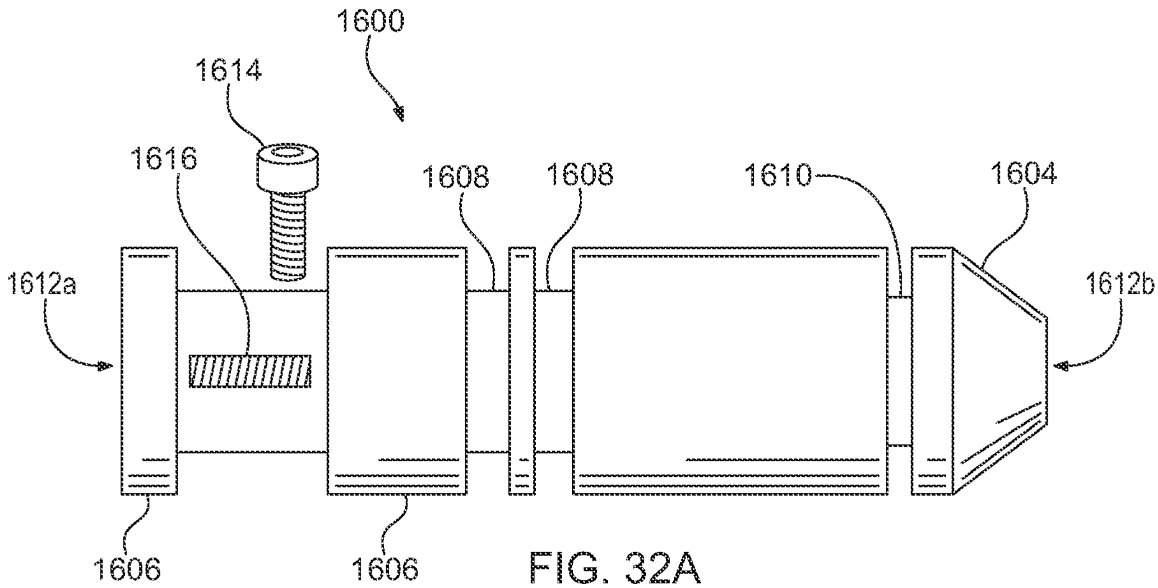

FIG. 32A

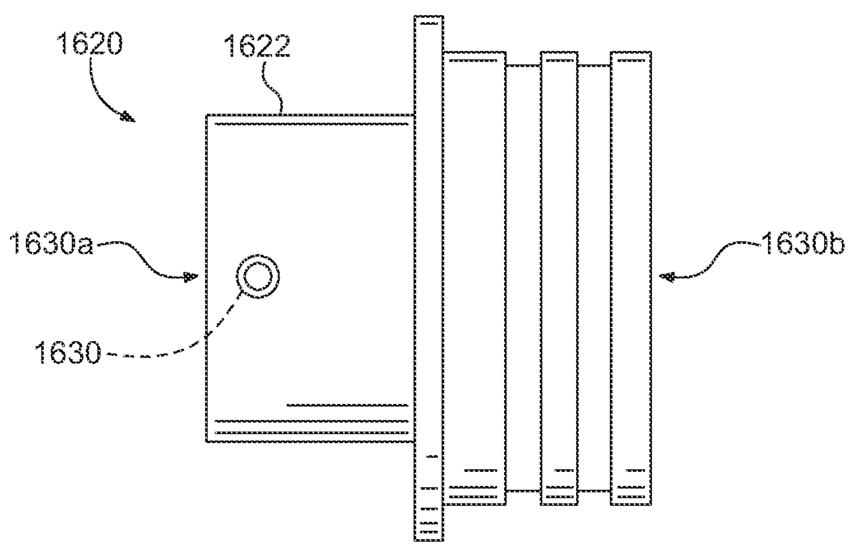

FIG. 32B

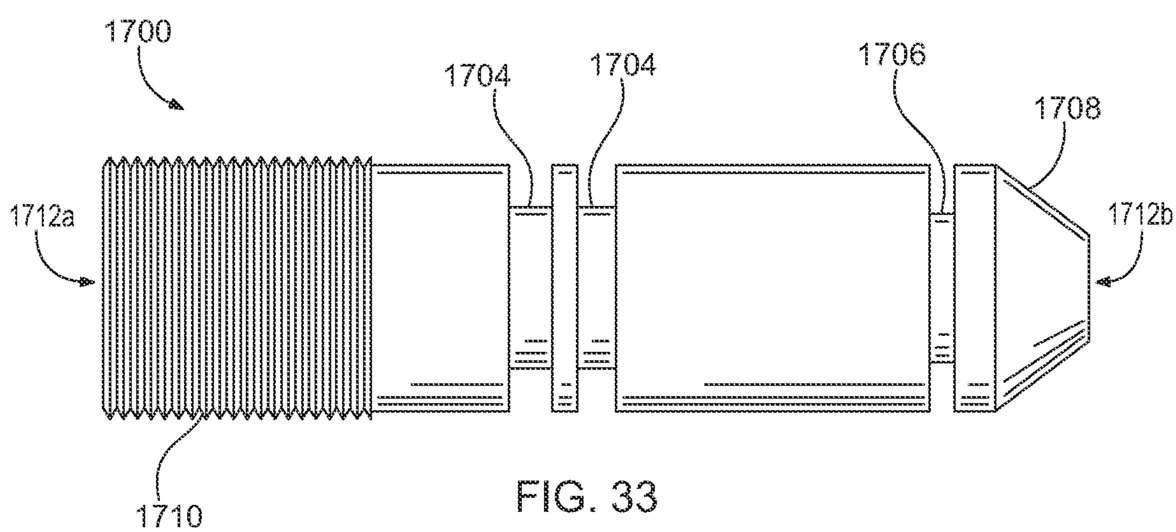

FIG. 33

DEVICES AND METHOD FOR REGULATING COOLER FLOW THROUGH AUTOMOTIVE TRANSMISSIONS

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation-in-part of U.S. patent application Ser. No. 17/288,720, filed on Apr. 26, 2021, which is a national stage entry of International Patent Application No. PCT/US2019/058831, filed on Oct. 30, 2019, which claims priority to U.S. Provisional Patent Application No. 62/752,539, filed on Oct. 30, 2018; and claims priority to U.S. Provisional Patent Application No. 63/458,465, filed on Apr. 11, 2023.

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

Not applicable.

THE NAMES OF THE PARTIES TO A JOINT RESEARCH AGREEMENT

Not applicable.

REFERENCE TO A "SEQUENCE LISTING," A TABLE, OR A COMPUTER PROGRAM LISTING APPENDIX SUBMITTED ON A COMPACT DISC

Not applicable.

STATEMENT REGARDING PRIOR DISCLOSURES BY THE INVENTOR OR A JOINT INVENTOR

Not applicable.

BACKGROUND OF THE INVENTION

Field of the Invention

The present invention relates to methods for improving OEM (original equipment manufacturer) systems for supplying cooled fluid lubricant through an automotive transmission, and to replacement parts for effecting said improvements, namely, improved thermal control system valves replacing OEM valves in automotive transmissions to improve performance and reduce maintenance costs.

Brief Discussion of the Prior Art

Most heat in an automatic transmission is generated in the torque converter (TC). Heat generation is relatively low during a lockup, or fluid coupling, phase, but during torque multiplication, and especially at maximum stall, high vortex flow forces fluid to make hard turns which generates a high level of fluid friction against internal component surfaces (for example, impeller, stator, and turbine). During sustained hard-working conditions in the transmission, fluid temperatures can flash up to 148.889-204.444° C. (300-400° F.). Thus, the most logical destination for liquid coolant, or fluid, flow is away from the torque converter is through a converter out line, which is a line directly away from the torque converter to the cooler. Since the fluid returning from the cooler is generally the coolest in the transmission, it is then

ideal for fluid to flow through lubrication circuits (LUBE), where it lubricates and cools intermeshing gears, washers, bearings, and bushings under load. From there, the fluid drains into the sump where it is drawn through the sump filter by the pump, which supplies that line pressure from which converter feed is derived. Thus, the basic cycle for this portion of transmission function is SUMP—FILTER—PUMP SUCTION—PUMP OUTPUT—LINE SUPPLY—TC FEED—TC OUT—COOLER—LUBE—back to SUMP. This strategy is typical and has been employed universally in all automatic transmissions for nearly a century with only a few rare exceptions.

In the same way a catalytic converter offers back pressure (i.e., resistance to flow) in an exhaust system, the transmission fluid cooler offers resistance in the fluid cooling system of the transmission, resulting in a pressure differential between the converter out line to the cooler (also referred to as an out-line) as compared to a cooler return line to lubricate the transmission (also referred to as an in-line). Two examples will suffice to illustrate this pressure differential. First, Honda 4 and 5 speed transmissions will typically flow 6.309E-5 m^3/s (1.5 gal/m (gallons per minute)) at 20-30 PSI (pressure/square inch) on the out-line, with about 41.369 kPa (6 PSI) in the in-line. A Ford 5R110W transmission will flow 0.0001262-0.0002524 m^3/s (2-4 GPM) at 137.895-275.79 kPa (20-40 PSI) through the out-line, with 68.948-103.421 kPa (10-15 PSI) on the cooler in-line. In this manner it is typical for most transmission cooler systems to maintain a 103.421-172.369 kPa (15-25 PSI) differential between out and in lines on either side of the cooler.

With an increased use of internal transmission computers, solenoids, sensors, pressure switches, and so forth, in modern automobiles, in combination with adaptive-learn and advanced shift control strategy programming, car manufactures have concluded that it is advantageous to warm the transmission fluid to an optimum operating temperature as quickly as possible, and thereafter maintain that controlled temperature throughout the drive cycle of the vehicle. The assumption is that if fluid temperature and viscosity are held constant, transmission functions can be controlled more consistently.

It has become quite fashionable in automotive engineering to employ the use of a thermostatic switch device to regulate flow through the transmission cooler. This "thermal element", as it is most commonly called, is placed somewhere in the thermal control system of the transmission where it can connect the out-line and the in-line circuits. In some cases, it is utilized in coordination with a flow control valve. Sometimes the thermal element itself is designed as a compound part, and functions as a thermally expanding valve. Other times, the thermal element itself is the flow stop device without the aid of a valve. In all cases, whether secondary devices are employed or not, the fundamental principles do not change. Fluid temperature is thermostatically controlled in similar fashions. ATF (automatic transmission fluid, or fluid) in transmissions has a preferred best working temperature of 148.889-204.444° C. (50-180° F.). Generally, optimum running temperature in transmissions has typically been 62.778-73.889° C. (145-165° F.). Leaks, valve sticking, and other high-temperature malfunctions tend to appear above 9.333° C. (200° F.), but most notably above 107.222-112.778° C. (225-235° F.) or right above the boiling temperature of water. That's where issues in the thermal flow-controlled transmissions typically arise. On the cold side, issues begin to arise below the freezing temperature of water, and at −37.222 to −40° C. (−35 to −40° F.) below zero ATF begins to gel.

The actual physical location of the thermostatic switch device, or thermal element, in principle can be anywhere these two circuits can be physically bridged. Further, the cooler, the out-line, the in-line, and thermostatic switch device together are often referred to as a thermal bypass system. To date, manufacturers have used five different locations for the thermal element:

TABLE 1

LOCATION	EXAMPLE TRANSMISSION
1. In Transmission Pump	Ford 5R110W
2. In Transmission Valve Body	Ford 4/5R55E, 5R55W, 5R55N, 5R55S
3. In Transmission Case Under Valve Body	Ford 6R80
4. In The Cooler	Dodge 68RFE, 545RFE
5. In The Cooler Lines	GM 6L80E; Ford 4R75W, 6F35

Regardless of location, the purpose of the thermal element is identical, and in many cases the same exact physical part is used, and by different manufactures. For example, one thermal element has been used in Ford, GM, Dodge, and Mercedes transmissions.

There is however a difference in accessibility and/or serviceability between these different locations. In the case of location #1, the transmission must be removed to access the pump. Location #2 requires valve body removal and disassembly. Location #3 requires valve body removal. Locations #4 and #5 are more easily and more cost effectively serviced, since they are external to the transmission. Thus, as more vehicles begin to use thermal flow control, location #5 is quickly becoming the preferred site for thermal element placement.

The structural shape of the thermal element also necessarily varies between most locations, most notability between a thermal element in the pump (location #1), an element in the valve body (location #2), an element in the case (location #3), and an element in the cooler or cooler lines (locations #4 and #5). Between the different possible locations, only valves used in locations #4 and #5 are likely to have an identical or highly similar structure, as the thermal element in the cooler lines (#4) can be integrally formed with the cooler (#5). Otherwise, a valve in the pump, for example, is not interchangeable with a valve meant to be used in a thermal element located in the transmission case.

There are a variety of different housings used to contain the thermal element of the same location between transmission manufacturers, but for the most part these are size and shape alterations necessary to accommodate differently sized cooler lines and different mounting locations. The valves used between these different housing shapes and sizes would be structured similarly, as the internal method of controlling cooler flow would be similar.

There are three possible states for known thermal bypass systems:

1. Fully OPEN when cooler is bypassed;
2. Fully CLOSED when ALL the flow is forced through the cooler; and
3. The INTERMEDIATE or PARTIAL ON state.

When the thermal element is fully open, fluid flows out of the converter, drops down and loops through the lockup control valve, and comes back up to a split. One direction goes to the flow valve. The other direction goes to the out fitting (out at the transmission and in at the cooler). Under pressure, flow always follows the path of least resistance, so the fluid flow chooses the in-line because the resistance in the cooler is much greater than that of the lube system. Pressure is transferred in both directions from the converter out circuit, but is equalized at the cooler return fitting, thus stopping cooler flow. Thus, in the OEM system, below a certain temperature, fluid flows in two directions and is stopped within the in-line near the connection of the cooler to the in-line. This system substantially prevents the flow of cooled fluid from the cooler to the transmission.

When the thermal element in an OEM system is fully closed, flow is restricted to one direction. This occurs when the fluid temperature is above the desired operating temperature. The thermal element is expanded sufficiently, due to silicon or a similar expanding element in the valve, to completely close the valve and prevent cooler bypass in order to force all fluid flow through the cooler to bring temperature down.

When the thermal element is cold, the valve allowing fluid flow through the thermal element is in an open, default position. When the thermal element is over the thermal temperature limit, for example, 121.111° C. (250° F.), the valve is in a closed, bypass position. But, as the fluid begins to cool, the thermal element begins to contract and holds the valve in a midway flow metering position where the valve is just cracking open in the bore. This is the normal operating state, where the element functions to sustain a predetermined ATF operating temperature, which is typically around 107.222-112.778° C. (225-235° F.). The thermal element holds the valve in a flow limiting position where part of the converter out flow goes through the cooler, and part of the flow bypasses through the thermal element directly to the transmission through the in-line. In this fashion, the fluid is partially cooled, and temperature is dynamically regulated. If ambient air temperature drops, and the cooler is more efficient, it bypasses more. If the air temperature rises, it pushes more fluid through the cooler.

There are multiple issues with this system, however, including:

- Overheating without setting diagnostic trouble codes as expected.
- Setting "phantom" codes as a result of erratic and inconsistent operation and/or temperature control.
- Silicon pack (thermal element valve) failure, leakage, and/or rupture with loss of fluid temperature control.
- Valves or other switching devices associated with the thermal element subject to sticking, which prevents proper and timely opening and closing of the thermal element.
- Cooler blocked and/or restricted with thermal system in cooler flow ON mode. This results in no cooler flow or lubrication, causing the planetary system to crash.
- When the thermal control system gets stuck in bypass mode, and cooler flow never begins, fluid can heat to nearly 400° F. At this temperature, if supplied oxygen, the ATF becomes a fuel and will sustain a fire.

Even when the thermal control system has not malfunctioned, high fluid operating temperatures increase expansion of valve body castings resulting in reduced and/or insufficient valve clearance. This causes slowed valve response to switching signals, sluggish regulation, and valve sticking with even the slightest amount of particle or carbon powder contamination. The same behavioral characteristics are observed with solenoids, especially PWM-type solenoids. Higher temperatures make it more difficult for adaptive learn solenoids to remain stable. The solenoids tend to drift, to dial in control of functions, while being compromised by temperature induced mechanical obstructions in multiple areas.

An improved thermal bypass control valve is discussed in U.S. Pat. No. 9,249,875 to Mason. The valve of Mason is designed to operate as part of a thermal element in location #2, the valve body, which is the location of the thermal element in Ford® 5R55 series transmissions. While the valve of Mason would be applicable to other transmissions with the thermal element located in location #2, the valve of Mason would not be applicable to transmissions having thermal elements in locations #1, #3, #4, or #5. Thus, the improved thermal bypass control valve provided in Mason transmission cannot be applied to all other transmissions.

In view of the foregoing, there is a need for an improved process for supplying cooler to all automotive transmissions to avoid potential damage that may be caused to transmissions due to failure or faulty operation of thermal elements across various models of transmissions. There is a need to ensure constant flow of fluid through the cooling system. There is also a need to ensure immediate fill of the cooling system with accurate fluid levels without a warm-up cycle. Further, there is a need for thermal bypass valves that accomplish these improvements across transmissions having the thermal element located in the pump, in the case, or in the cooler or cooler lines.

BRIEF SUMMARY OF THE INVENTION

The present invention is directed toward improved methods for regulating thermal control systems in automotive transmissions and thermal bypass valves that replace factory-original thermal bypass valves in thermal elements located at different locations within transmissions across multiple makes and models of such transmissions.

It is an objective of the instant disclosure to provide a method of converting an OEM thermal bypass control system from a three-state system to a two state system, comprising: removing a three-state OEM valve from a thermal element, wherein the OEM valve prevents any fluid flow between a transmission and a cooler in a default state, allows a full fluid flow between the transmission and the cooler past a fluid temperature threshold, and allows a partial fluid flow along a fluid temperature range below the fluid temperature threshold; replacing the OEM valve with a two-state valve in the thermal element; and the two-state valve allowing full fluid flow between the transmission and the cooler in a default state, and allowing for fluid bypass of the cooler when a pressure differential between fluid flowing from the transmission to the cooler and fluid flowing from the cooler to transmission exceeds a predetermined range.

Another objective of the instant disclosure is to provide a thermal bypass valve for installation into a transmission pump, the valve having a cylindrical blocker valve having a groove extending around a circumference of the blocker valve and along a length of the blocker valve, the groove defining an upper valve portion and a lower valve portion; a relief valve having a uniformly cylindrical body, a cylindrical cavity extending within and along a partial length of the relief valve, wherein the cylindrical cavity is open at a free end of the relief valve, and a cylindrical member extending from the relief valve along an end of the relief valve opposite the free end, wherein the cylindrical member has a smaller diameter than the relief valve; a spring; a valve plug; and a clip.

Another objective of the instant disclosure is to provide a thermal bypass valve for installation into a transmission case, the thermal bypass valve having a cylindrical body having an inner cylindrical cavity extending along a longitudinal length of the cylindrical body, the cylindrical body having a free end contiguous with the cylindrical cavity; a raised band coaxially extending around a circumference of the cylindrical body and proximate to the free end; a second raised band coaxially extending around a circumference of the cylindrical body along an end opposite the free end; a coaxial protrusion extending from the end opposite the free end; a tube extending within the cylindrical cavity proximate to the free end and perpendicularly to the longitudinal length of the cylindrical body, wherein the tube extends through opposite sides of the cylindrical body through two holes in the cylindrical body; a relief valve slidable within the cylindrical cavity; and a spring within the cylindrical cavity and compressible between the relief valve and a closed end of the cylindrical cavity; wherein a plurality of openings in the cylindrical body are positioned adjacent to the raised band, and a relief opening in the cylindrical body is positioned adjacent the second raised band; and wherein the thermal bypass valve is configured to insert into a transmission case of an automotive transmission.

Yet another objective of the instant disclosure is to provide a thermal bypass valve for installation into a cooler or external block connected to cooler lines, the thermal bypass valve having a cylindrical body, an inner cylindrical cavity extending along a longitudinal length of the cylindrical body, the cylindrical body having a free end contiguous with the cylindrical cavity, wherein a second cylindrical cavity is contiguous with the cylindrical cavity opposite the free end; a grooved blocker portion attached to the cylindrical body opposite the free end and adjacent to a grooved portion of the cylindrical cavity, the grooved portion having a plurality of openings into the second cylindrical cavity, wherein the grooved blocker portion has two grooves each extending along a circumference of the grooved blocker portion; and, a cap portion attached to the grooved blocker portion at an end opposite of the cylindrical body, the cap having a larger diameter than the grooved blocker portion, and a having a member attached at an end opposite the grooved blocker portion, a piston having a first cylindrical portion adjacent to a second cylindrical portion, the first cylindrical portion and second cylindrical portion defining a central cylindrical cavity extending along a longitudinal length of the piston and open at opposing free ends of the piston, wherein the first cylindrical portion has a greater diameter than the second cylindrical portion; a spring having an end with a smaller diameter than an opposing end; a bearing ball; two large O-rings; and a small O-ring, wherein the bearing ball is configured to rest partially within the second cylindrical cavity and secured by the spring along the end with a smaller diameter, and the spring is compressible by the piston along a free end of the first cylindrical portion, wherein the piston slidably engages the inner cylindrical cavity of the sleeve along the first cylindrical portion, wherein each O-ring of the two large O-rings engages a groove of the two grooves of the grooved blocker portion, wherein the small O-ring engages the circumference of the second cylindrical portion of the piston adjacent to the first cylindrical portion, and wherein the thermal bypass valve is configured to insert into an external thermostat block of an automotive transmission.

Embodiments of the present invention are intended to overcome the problems associated with lockup or other damage to components and gears within a transmission, especially the overdrive sun and planet gears of transmissions, by providing a replacement thermal bypass valve system which replaces the OEM thermal bypass valve system to provide a continuous flow cooler.

Another objective of the present invention is to provide replacement thermal bypass valves for different transmission series that default to a cooler flow ON state, instead of a cooler flow OFF state which is present in OEM thermal bypass valve systems.

A further objective of the present invention is to provide replacement thermal bypass valves for different transmission series that do not require a preheat cycle to raise temperatures to +212° F./+100° C. before allowing cooler to flow.

Yet another object of the present invention is to provide improved thermal control systems to increase cooler supply to and within transmissions to improve overall performance of the transmissions.

The present invention also provides for a thermal bypass valve, having a plug having an inner cavity longitudinally extending between two oppositely oriented openings and a cap configured to prevent fluid flow between the thermal bypass valve and a valve bore in a transmission, the plug configured to insert into an opening of the valve bore; a longitudinally extending plunger configured to be secured coaxially through the inner cavity of the plug and extending beyond the two oppositely oriented openings of the plug; a blocker valve having and a fluid channel longitudinally extending between two oppositely oriented blocker valve openings; a first spring configured to be secured between the plunger and the blocker valve; and a second spring configured to be secured between the blocker valve and a surface of the valve bore, wherein, when the thermal bypass valve is installed in a transmission, a position of the plunger relative to the plug is manually changeable to enable full transmission fluid flow to a cooler, full transmission fluid bypass of the cooler, and mixed transmission fluid flow to the cooler and to a warmer.

The thermal bypass valve may include a flange secured to the plunger adjacent to a first end of the plunger, the flange configured to engage the first spring and extending perpendicularly from the plunger relative to a longitudinal axis of the plunger.

Further, the blocker valve may have a lip for engaging the first spring.

The plug may further include an upper member and a lower member, the upper member and lower member longitudinally extending on either side of the cap.

The lower member may have one or more grooves extending circumferentially along an outer surface of the lower member, and may further include one or more plug O-rings, one plug O-ring of the one or more plug O-rings corresponding to a single groove of the one or more grooves extending along the outer surface of the lower member.

Further, the plunger may have one or more grooves extending circumferentially along an outer surface of the plunger, and may further include one or more plunger O-rings, one plunger O-ring of the one or more plunger O-rings corresponding to a single groove of the one or more grooves extending circumferentially along the outer surface of the plunger.

Further, the plunger may have a tapered surface along the first end.

The thermal bypass valve may further include a retaining clip, wherein the plug has a retaining clip groove extending circumferentially along an outer surface of the upper member, wherein the plunger has a plurality of flow grooves extending circumferentially along an outer surface of the plunger, and wherein the retaining clip is configured to simultaneously secure in the retaining groove and one flow groove of the plurality of flow grooves.

The thermal bypass valve may further a means to manually change a position of the plunger relative to the plug when the plunger is secured coaxially through the inner cavity of the plug and extending beyond the two oppositely oriented openings of the plug.

A better understanding of the systems and methods will be had with reference to the several views of the drawings described herein.

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWING

The present invention is shown and described in the following drawings:

FIG. **1** shows components of a factory-original thermal control bypass valve for installation in a transmission pump;

FIG. **2** shows a thermal control bypass valve embodiment of the present invention and constituent components designed for installation in a transmission pump;

FIG. **3** shows a side view of a relief valve of the thermal control bypass valve embodiment, as provided in FIG. **2**;

FIG. **4** shows a side view of a blocker valve of the thermal control bypass valve embodiment, as provided in FIG. **2**;

FIG. **5** shows installation of the thermal control bypass valve embodiment of FIG. **2** into a transmission pump;

FIG. **6** shows a factory-original wax pellet style thermal control bypass valve for installation in a transmission case;

FIG. **7** shows a thermal control bypass valve embodiment of the present invention designed for installation in a transmission case;

FIG. **8** shows a cross-sectional side view of the thermal control bypass valve embodiment, as shown in FIG. **7**, without an inner spring and valve;

FIG. **9** shows a cross-sectional side view of the thermal control bypass valve embodiment, as shown in FIG. **7**, with the inner spring and valve;

FIG. **10** shows a cross-sectional side view of the thermal control bypass valve embodiment rotated 90° along a longitudinal axis relative to the view of FIG. **9**;

FIG. **11** shows a side view of the thermal control bypass valve embodiment rotated 180° along a longitudinal axis relative to the view of FIG. **9**, and without a cross-sectional view;

FIG. **12** shows a bottom view of the thermal control bypass valve embodiment of FIG. **7**;

FIG. **13** shows a top view of the thermal control bypass valve embodiment of FIG. **7**;

FIG. **14** shows removal of the factory-original wax pellet style thermal control bypass valve of FIG. **6** from a transmission case and replacement with the thermal control bypass valve embodiment of FIG. **7**;

FIG. **15** shows a thermal control bypass valve embodiment of the present invention designed for installation in a thermal element block in cooler lines;

FIG. **16**A shows a cross-sectional side view of a sleeve and ball of the thermal control bypass valve embodiment of FIG. **15**;

FIG. **16**B shows a side view of the sleeve of FIG. **16**A rotated 45° along a central longitudinal axis;

FIG. **17** shows a cross-sectional side view of a sleeve of the thermal control bypass valve embodiment of FIG. **15**;

FIG. **18** shows a cross-sectional side view of the thermal control bypass valve embodiment of FIG. **15** with constituent parts operably assembled together;

FIG. **19** shows installation of the thermal control bypass valve embodiment of FIG. **15** into factory-original thermal element block;

FIG. 20A shows a factory-original thermal control bypass valve in a default position to allow fluid bypass;

FIG. 20B shows an illustration of a fluid circuit through a thermostat block with the valve of FIG. 20A in a default, cooler bypass position according to prior art systems;

FIG. 21A shows a factory-original thermal control bypass valve in an altered, extended position to allow fluid flow to the cooler;

FIG. 21B shows an illustration of a fluid circuit through a thermostat block with the valve of FIG. 21A in an altered position to permit fluid flow through the cooler according to prior art systems;

FIG. 22 shows an illustration of a fluid circuit through a thermal element with a valve in a default position allowing fluid flow through the cooler according to an embodiment of the present invention;

FIG. 23 shows an illustration of a fluid circuit through a thermal element with the valve in an altered position to allow cooler bypass according to an embodiment of the present invention;

FIG. 24 shows an illustration of a fluid circuit through a transmission pump according to an embodiment of the present invention;

FIG. 25 shows a thermal control bypass valve embodiment of the present invention designed for installation in a thermal element located in the cooler;

FIG. 26A is a side view of a transmission with arrows showing transmission fluid flow;

FIG. 26B is an opposite side view of the transmission shown in FIG. 26A along with an engine coolant warmer that is attached to the transmission along corresponding openings;

FIG. 26C is a perspective view of the engine coolant warmer shown in FIG. 26B;

FIG. 26D is a side view of a prior art OEM thermal bypass device corresponding to the transmission of FIG. 26A;

FIG. 26E is a front view of the transmission shown in FIG. 26A showing an opening corresponding to installation of the prior art OEM thermal bypass device of FIG. 26E.

FIG. 27 is a side view of a thermal bypass device according to an embodiment of the present invention;

FIG. 28 is a side view of the transmission of FIG. 26A, where the OEM thermal bypass device of FIG. 26D is replaced with the thermal bypass device of FIG. 27;

FIG. 29 is a diagram along the side view of the transmission of FIG. 26A and the thermal bypass device of FIG. 27 showing the thermal bypass device installed within the transmission in three different configurations and a corresponding position of a cylindrical blocker valve of the thermal bypass device in each different configuration;

FIG. 30A is a side view of a cap portion of the thermal bypass device of FIG. 27;

FIG. 30B is a bottom view of the cap portion shown in FIG. 30A;

FIG. 30C is a side view of a plunger of the thermal bypass device of FIG. 27;

FIG. 30D is a cross-sectional view of the blocker valve of the thermal bypass device shown in FIG. 27;

FIG. 31 is an alternate embodiment of the plunger with a threaded diameter corresponding to a screw for adjustment of the plunger relative to the cap portion;

FIG. 32A is an alternate embodiment of the plunger with a threaded groove corresponding to a screw for adjustment of the plunger relative to the cap portion;

FIG. 32B is an alternative embodiment of the cap portion with a hole corresponding to the screw of FIG. 32A; and

FIG. 33 is an alternate embodiment of the plunger with a threaded end corresponding to a nut for adjustment of the plunger relative to the cap portion.

DETAILED DESCRIPTION OF THE INVENTION

In the following detailed description, systems, apparatuses, and methods for improving thermal control in automotive transmissions are described by providing references to the accompanying drawings which form a part of the description of how the invention works and does not limit the scope of the present invention. The present invention solves the problem of insufficient cooler flow to and through automotive transmissions due to OEM parts set to bypass fluid flow through the cooler by providing thermal bypass valves with improved structures that replace OEM thermostatic valves and increase cooler flow to and through referenced transmissions by defaulting the thermal control system to fluid flow through the cooler.

It will be appreciated that for simplicity and clarity of illustration, reference numerals may be repeated among the figures to indicate corresponding or analogous elements. In addition, numerous specific details are set forth to provide a thorough understanding of the embodiments described herein. However, it will be understood by those of ordinary skill in the art that the embodiments described herein may be practiced without these specific details. In other instances, well-known methods, procedures, and components have not been described in detail so as not to obscure the embodiments described herein. Also, the description is not to be considered as limiting the scope of the embodiments described herein.

Dimensions are provided for valves and their individual parts and components. Such dimensions are typically identified as either diameter or length and denoted with D_n or L_n. The dimensions are specific to the part particularly referenced and do not share common values across like-numbered elements in other valves or valve parts.

An OEM thermostatic valve **100** for installation in a transmission pump is shown in FIG. **1**. The valve **100** includes an upper portion **102**, a lower portion **104**, a spring **106**, a valve plug **110**, and a clip **108** to secure the valve plug in place. As discussed, the OEM thermostatic valve **100** defaults to a cooler flow OFF state, preventing flow of cooled fluid through the cooler system. In the OFF state, fluid is allowed to flow in both directions to the cooler (i.e. clockwise and counterclockwise in a fluid circuit), which causes a near equilibrium that substantially stops flow out of the cooler near a connection of the cooler and in-line. To switch the OEM thermal control system to a cooler flow ON state, a preheat cycle is required to raise temperatures to above 100° C. (212° F.). This in turn causes the OEM thermostatic valve to begin to close off fluid flow both directions to the cooler, and force fluid flow in a single direction such that hot fluid leaves the transmission enters the cooler to be cooled and is returned as cool fluid to the transmission. When the transmission is cooler than 165° F. (nominal) the flow of lubricant to the cooler is cut by 90%. The reduced flow is directed to lube, then to the pan. Therefore, if the transmission is cool enough for the thermostat to remain in the cooler flow OFF state, 90% of the new fluid mixes with the old fluid, while only 10% flows out and is replaced. If the OEM thermal bypass system malfunctions and the system is not turned to an ON state, the oil and transmission overheat due to a lack of freshly cooled lubricant supply. This in turn affects the pump, valve body, solenoids, sealing rings, and clutch drum operation.

FIG. **2** illustrates an embodiment of an improved thermal bypass valve **200** for installation in the transmission pump. A preferred embodiment of the thermal bypass valve **200** includes a blocker valve **204**, a relief valve **202**, and spring **206**. The thermal bypass valve **200** reuses the valve plug **110** and clip **108** of the OEM thermostatic valve **100** in this embodiment. A new plug **110** and clip **108** may be provided in further embodiments, but reusing the plug and clip from the OEM valve saves on costs associated with converting the thermal cooling system according to embodiments of the current invention. The remaining components of the thermostatic valve **100**, namely the upper portion **102**, lower portion **104**, and spring **106**, are removed from the transmission oil pump and discarded or recycled.

FIGS. **3** and **4** provide descriptions and dimensions of the relief valve **202** and blocker valve **204**. Dimensions provided for the valves **202** and **204** are preferred for a bypass valve **200** that is installed in the transmission pump of a Ford® 5R110W series transmission. The dimensions may be altered to conform to specific dimensions, if different, of other transmissions having the thermal element and corresponding thermal bypass valve in the transmission pump.

Referring now to FIG. **3**, a side view of the relief valve **202** of the thermal bypass valve **200** is shown. The relief valve **202** includes a uniformly cylindrical body **208** having a length L_1 and a diameter D_1. In the preferred embodiment, L_1 is 24.13 mm (0.950") and D_1 is 12.085 mm+/−0.0508 mm (0.4758"+/−0.002"). A cylindrical cavity **212** having a uniform diameter D_2 extends within and along a partial length L_2 of the cylindrical body **208**. The cavity **212** is contiguous with and open at a free end **209** of the cylindrical body **208** and is closed at an end **211** of the cavity opposite the free end. In the preferred embodiment, L_2 is 11.176 mm (0.440") and D_2 is 8.382 mm (0.330"). A cylindrical member **210**, having a length L_3 and a diameter D_3, is coaxial with and extends away from the cylindrical body **208** along an end **213** of the cylindrical body opposite the free end **209**. In the preferred embodiment, L_3 is 3.810 mm (0.150") and D_3 is 2.540 mm (0.100").

Referring now to FIG. **4**, a side view of the cylindrical blocker valve **204** of the thermal bypass valve **200** is shown. The cylindrical blocker valve **204** includes a groove **220** having a length L_3 and a diameter D_3, In the preferred embodiment, L_3 is 5.334 mm (0.210") and D_3 is 10.033 mm (0.395"). The groove **220** defines an upper cylindrical portion **222** and a lower cylindrical portion **224** directly adjacent to and on either side of the groove. The upper cylindrical portion **222** has a length L_2 and a diameter D_1. In the preferred embodiment, L_2 is 12.700 mm (0.500") and D_1 is 12.073 mm+/−0.0508 mm (0.4753"+/−0.002"). A beveled edge **226** of the upper cylindrical portion **222** is adjacent to the groove **220**, and has a length L_5, included within L_2, and decreases in diameter from D_1 to a diameter D_2, directly adjacent to the groove **220**, In the preferred embodiment, L_5 is 2.159 mm (0.085") and D_2 is 11.303 mm (0.445"). The lower cylindrical portion **224** has a length L_4 and a diameter D_4, In the preferred embodiment, L_4 is 17.526 mm (0.690") and D_4 is 11.3157 mm+/−0.00508 mm (0.4455"+/−0.0002"). The lower cylindrical portion **224** also has a beveled edge **228** along a free end **230** of the lower cylindrical portion **224**. The beveled edge **228** has a length L_6 and decreases from D_4 into a diameter D_5. In the preferred embodiment, L_6 is 2.159 mm (0.085") and D_5 is 10.541 mm (0.415"). The cylindrical blocker valve has an overall length L_1, which is the sum of L_2, L_3, and L_4, which is 35.56 mm (1.400") in the preferred embodiment.

FIG. **5** shows the thermal bypass valve **200** being inserted into a representative transmission oil pump **201**. To install the preferred embodiment of the thermal bypass valve **200**, the clip **108** is first removed from the valve plug **110**. The valve plug **110** is then removed, and both the valve plug and clip **108** are saved. Next, the upper portion **102**, lower portion **104**, and spring **106** of the OEM thermostatic valve **100** are removed from the cooler flow control valve section of the oil pump **201**. The upper portion **102**, lower portion **104**, and spring **106** of the OEM thermostatic valve **100** may then be discarded or recycled.

After discarding the OEM thermostatic valve **100**, the thermal bypass valve **200** of the present invention may then be installed. First, the cylindrical blocker valve **204** is inserted into cooler flow control valve section of the oil pump **201** along the free end **230** of the lower cylindrical portion **224** having the beveled edge **228**. The relief valve **202** is then inserted such that the cylindrical member **210** rests along a free end **232** of the upper cylindrical portion **222** of the blocker valve **204**. The spring **206** is then inserted such that a portion of the spring's length is contained within the cylindrical cavity **212** of the relief valve **202**. The spring **206** has a diameter that allows it to be secured within the cylindrical cavity **212** of the relief valve **202** without substantial lateral movement that would otherwise contribute to wearing of the cavity or misalignment of the valves **202** and **204** in the pump **201**. The valve plug **110** is then once again secured to the oil pump **201** and is further secured by the clip **108** to keep the thermal bypass valve **200** in the oil pump.

Further discussion of the cooler circuit of the thermal control system through the pump upon replacing the OEM valve **100** with the thermal bypass valve **200** or a similar embodiment is provided in reference to FIG. **24**.

In a default state, or position at vehicle start-up, the relief valve **202** blocks a line connection path between the in-line and out-line through the thermal element. This forces cooler fluid to flow from other transmission systems to the cooler via the out-line and return from the cooler to the transmission systems via the in-line. If PSI within the fluid circuit of in-lines and out-lines reaches the blow-off PSI, the relief valve **202** is forced outward away from a center of the pump, towards the plug **110**, and the spring **206** is compressed, thus allowing for cooler fluid flow through the line connection path and to include the thermal element in the fluid circuit to bypass flow to the cooler **1122**. This scenario would occur if a blockage in the cooler **1122** prevents fluid flow through the cooler. When pressure decreases in the out-lines, forces on the relief valve **202** decrease and the spring **206** expands to push the relief valve back towards the blocker valve **204**. This re-establishes flow from the pump and transmission to the cooler. The spring **206** is in a vented area between the relief valve **202** and plug **110**, so there is no counterbalance oil pressure on the spring side of the relief valve. Therefore, the blow-off PSI is equal to the spring tension divided by the area of the end of the valve **202**.

An OEM wax pellet style thermostatic valve **400** for insertion into a transmission case, or casing, is shown in FIG. **6**. By way of example, the thermostatic valve of the Ford® 6R60, 6R75, and 6R80 series transmissions are housed in the transmission case and are typically used in Ford® Explorer models (6R60 series transmission), Ford® Expedition and Lincoln® Navigator models (6R75 series transmission), and Ford® Mustang and F150 models (6R80 series transmission). The OEM thermostatic valve **400** has sleeve **402**, brass pellet **404**, and spring **406**, which are inserted into a transmission case or casing. As with the OEM thermostatic valve **100**, the OEM thermostatic valve **400** defaults to a cooler flow OFF state.

FIG. **7** illustrates an embodiment of a thermal bypass valve **300** for insertion into a transmission casing. Dimensions for the thermal bypass valve **300** are provided by way of example to fit into the corresponding thermal element for the Ford® 6R60, 6R75, and 6R80 series transmissions to replace the OEM valve **400**. However, the dimensions may be altered to fit within other transmissions having the thermal element located within the transmission case.

Referring now to FIG. **8**, a side view of an embodiment of the thermal bypass valve **300** is provided. The thermal bypass valve **300** has a cylindrical body **304**. The cylindrical body **304** has an inner cylindrical cavity **312** extending along a longitudinal length of the cylindrical body, the cylindrical body having a free end **310** contiguous with the cylindrical cavity, and along which the cylindrical cavity is open. The cylindrical cavity **312** has a length L_9 and a diameter D_4. In a preferred embodiment of the thermal bypass valve **300**, L_9 is 31.75 mm (1.250") and D_4 is 10.01268 mm+/−0.00508 mm (0.3942"+/−0.0002"). A raised band **308** coaxially extends around a circumference of the cylindrical body **304** and is located proximate to the free end **310**. A second raised band **306** coaxially extends around a circumference of the cylindrical body **304** along an end **307** opposite the free end **310**. The raised band **308** and second raised band **306** have a length L_5 and a diameter D_3. In a preferred embodiment of the thermal bypass valve **300**, L_5 is 3.81 mm (0.150") and D_3 is 17.28724 mm+/−0.00508 mm (0.6806"+/−0.0002"). A portion of the cylindrical body between the raised band **308** and second raised band has a length L_3 and a diameter D_1. In a preferred embodiment of the thermal bypass valve **300**, L_3 is 21.082 mm (0.830") and D_1 is 14.732 mm (0.580"). Another portion of the cylindrical body **304** between the raised band **308** and free end **310** has a length L_4 and has the diameter D_1. In a preferred embodiment of the thermal bypass valve **300**, L_4 is 4.318 mm (0.170"). A coaxial cylindrical protrusion **302** extends from the end **307** opposite the free end **310**. The cylindrical protrusion **302** has a length L_2 and a diameter D_2. In a preferred embodiment of the thermal bypass valve **300**, L_2 is 17.78 mm (0.700") and D_2 is 11.43 mm (0.450"). A relief opening **318** is provided in the cylindrical body **304** and is positioned between the raised band **308** and second raised band **306**, and proximate to the second raised band. In a preferred embodiment of the thermal bypass valve **300**, the relief opening **318** has a diameter of 1.5875 mm (0.0625") and a center of the relief opening is positioned 28.194 mm (1.110") away from the free end **310**. A plurality of openings **316** are provided in the cylindrical body **304** between the raised band **308** and second raised band **306**, and proximate to the raised band. In a preferred embodiment of the thermal bypass valve **300**, each opening **316** has a diameter of 3.175 mm (0.125") and is positioned 9.906 mm (0.390") away from the free end **310**. In a preferred embodiment of the thermal bypass valve **300**, four openings **316** are provided proximate to the raised band and evenly spaced along the circumference of the cylindrical body **304**. In a preferred embodiment of the thermal bypass valve **300**, the entire length L_1 of the valve is 50.800 mm (2.000").

Referring now to FIG. **9**, a side view of the preferred embodiment of the thermal bypass valve **300** is shown including a cylindrical relief valve **320** slidable within the cylindrical cavity **312**. A spring **322** is also positioned within the cylindrical cavity **312** between the relief valve **320** and a closed end **309** of the cylindrical cavity. The relief valve **320** has a cylindrical member **324** coaxial with the relief valve and extending from a free end **321** of the relief valve. The spring **322** is semi-compressed within the cylindrical cavity **312** to keep the member **324** in contact with a tube **326** extending across the cavity. During operation, forces may act on the relief valve **320** forcing the relief valve to slide further toward the closed end **309** of the cavity **312**, compressing the spring **322** further. When such forces are absent, the spring **322** expands to slide the relief valve **320** back to its relaxed, or default, position, with the member **324** in contact with the tube **326**, once the counteracting forces are no longer present. The relief valve **320** and spring **322** are omitted from the view provided in FIG. **8** for the purpose of clearly showing the varying lengths and diameters of the various elements of the valve **300**. FIG. **9** shows a view of the structures of the valve **300** within the cavity **312** omitted in FIG. **8**.

FIG. **10** provides a side view of the thermal bypass valve **300** rotated 90° relative to the view provided in FIGS. **8** and **9**. The tube **326** extends across the cylindrical cavity **312** between the free end **310** and the raised band **308**, and perpendicularly to the longitudinal length of the cylindrical body **304**. The tube **326** extends within two holes **314** in the cylindrical body **304** proximate to the free end **310**, and the two holes **314** are positioned 180° relative to each other along the circumference of the cylindrical body **304**. In a preferred embodiment of the thermal bypass valve **300**, each of the two holes **314** are 1.5875 mm (0.0625") in diameter and a center of each hole is positioned 2.032 mm (0.080") from the free end **310**. Further, the tube is preferably hollow such that the two holes **314** and tube **326** form an enclosed cylindrical space across the cylindrical cavity **312**. However, the tube **326** may be solid in other embodiments of the valve **300**. A view from the opposite side of the valve **300** mirrors the view shown in FIG. **10**

FIG. **11** provides a side view of the thermal bypass valve **300** rotated 180° relative to the view provided in FIGS. **8** and **9**. In this view, one of the two holes **314** is shown, along with an opening of the plurality of openings **316**. As there is only one relief opening **318**, and it is on the opposite side of the cylindrical body **304**, another relief opening is not shown.

FIG. **12** shows a bottom view of the thermal bypass valve **300** provided in FIGS. **8** and **9**. The raised band **308** sets the outer most diameter of the valve **300**, along with the second raised band **306**. The two holes **314** and tube **326** are shown from a better view. The member **324** of the relief valve **320** rests against the tube **326**, keeping the relief valve within the cylindrical cavity **314**.

FIG. **13** shows a top view of the thermal bypass valve **300** provided in FIGS. **8** and **9**. This view would be seen by a user upon installing the valve **300** into the transmission casing. The second raised band **306**, along with the raised band **308**, sets the outer most diameter for the thermal bypass valve **300**. The cylindrical protrusion **302** is shown having a smaller diameter than the second raised band **306**, and the remaining elements are obscured in this view.

FIG. **14** demonstrates how the thermal bypass valve **300** replaces the OEM thermostatic valve **400** in transmissions having the thermal element in the transmission casing. After locating the thermostatic valve opening on the transmission case, the OEM thermostatic valve **400** is removed, including the sleeve, wax pellet, and spring. The OEM thermostatic valve **400** may then be discarded or recycled. The thermal bypass valve is then inserted into the thermostatic valve opening, such that the free **310** is inserted first and the cylindrical protrusion **302** is still visible after insertion.

In operation, the free end **310** of the valve **300** would be inserted into an opening between a thermal element chamber and an out-line leading fluid from the transmission to the cooler. Alternately, the valve sits in the thermal element chamber such that the raised band **308** creates a seal between a connection point of the in-line and out-line, such that fluid must flow from the out-line through free end **310** and through the cavity **312** to reach the in-line. In the default position, shown in FIG. **9**, the relief valve **320** blocks access to the thermal element chamber and in-line from the out-line, preventing cooler bypass, and forcing fluid to travel from the transmission through the out-line to the cooler and back to the transmission via the in-line, thereby ensuring cool fluid return to the transmission. Under high pressure in the out-line, such as when a blockage in the cooler prevents flow out through the in-line, forces on the relief valve **320** compress the spring **322** moving the relief valve into the cavity and toward end **309**. Once the spring **322** is compressed far enough, openings of the plurality of openings **316** allow fluid to exit the cavity **322** and enter the in-line to the transmission. This allows the fluid to bypass the cooler. Relief opening **318** allows fluid and/or air to escape or enter the cavity **312** between the relief valve **320** and end **309** during compression or expansion of the spring **322**. Raised band **306** forms the outer diameter and seal between the thermal element chamber, in-line, and external environment, closing the fluid circuit at the thermal element location.

An embodiment of a thermal bypass valve **600** for transmissions having a thermal element external to the transmission, in either a block in the cooler in-line and out-line or in the cooler itself, is shown in FIG. **15**. Dimensions provided in referenced to the valve **600** pertain specifically to an improved valve fitting into an external thermostat block of the Ford® 6R80E series transmission, model year 2014 and up, that is removably attached to an outside of the transmission case. However, the dimensions of the valve may be changed to fit other transmissions having the thermal element within the cooler or external blocks in the cooler line.

FIG. **15** shows the individual components of the thermal bypass valve **600** that fit into the external thermostat block after removal of the OEM thermostatic valve, including a sleeve **602**, two O-rings **604** and **606**, a conical spring **608**, a bearing ball **610**, a piston **614**, and a piston O-ring **612**. Further discussion of these components is provided below.

Referring now to FIG. **16**A, the sleeve **602** of the thermal bypass valve **600** is shown in greater detail, along with the bearing ball **610**. The sleeve **602** has a cylindrical body **620** with a cylindrical cavity **626** extending along a longitudinal length of the cylindrical body. The cylindrical cavity **626** is contiguous with and open along a free end **621** of the cylindrical body **620**. An opposite inner end **623** of the cylindrical cavity **626** is beveled toward a second cylindrical cavity **628**, which in turn is contiguous with the beveled inner end **623** of the cylindrical cavity **626**. The beveled inner end **623** is preferred, although not necessary, with use with a bearing ball **610**. The beveled end **623** helps to ensure that the bearing ball **623** sits fully within the opening between the cylindrical cavities **626** and **628** between expansion and contraction of the spring **608**. A squared end runs the risk of the bearing ball **610** getting caught between the spring **608** and end **623** without fully blocking fluid access to the cylindrical cavity **626**. The beveled edge ensures that, even in the event of a misalignment, the bearing ball **610** returns to fully sit within the opening between the cavities **626** and **628** to prevent and/or restrict fluid flow to the cylindrical cavity **626**.

The second cylindrical cavity **628** has a smaller diameter and length than the cylindrical cavity **626**. A closed end **625** of the second cylindrical cavity is pitched toward a grooved blocker portion **622** of the sleeve **602**. The cylindrical body **620** has a groove **629** extending around an outer circumference of the cylindrical body **620** along an end **627** opposite the free end **621**. A plurality of openings **630** in the groove **629** lead into the second cylindrical cavity **628**. The cylindrical body **620**, including the groove **629**, has a length L_1. In a preferred embodiment of the valve **600**, L_1 is 26.543 mm (1.045"). The cylindrical body **620**, not including the groove **629**, has a length L_5. In a preferred embodiment of the valve **600**, L_5 is 23.241 mm (0.915"). The cylindrical body **620** has a diameter D_7. In a preferred embodiment of the valve **600**, D_7 is 18.4658 mm (0.727"). The cylindrical cavity **626** has a diameter D_6. In a preferred embodiment of the valve **600**, D_6 is 18.4658 mm (0.5469"). The second cylindrical cavity **628** has a diameter D_5. In a preferred embodiment of the valve **600**, D_5 is 6.9088 mm (0.272"). Each of openings of the plurality of openings **630** has a diameter of 3.175 mm (0.125"). In a preferred embodiment of the valve **600**, there are four openings **630** spaced equidistantly around the circumference of the groove **629**. There is a 0.127 mm (0.005") clearance between each opening of the plurality of openings **630** and the grooved blocker portion **622**.

Still referring to FIG. **16**A, the grooved blocker portion **622** of the sleeve **602** is contiguous at an end with the groove **629** and end **627** of the cylindrical body **620**. The grooved blocker portion **622** is itself cylindrical and has two grooves **632** and **633**. The grooves **632** and **633** are evenly spaced relative to each other about the circumference of the grooved blocker portion **622**. A cap **634** is attached to the grooved blocker portion along an end opposite the cylindrical body **620**. In use, the cap **634** seals the cooler and other components of the thermal bypass valve **600** within the external thermostat block. A cylindrical member **624** is attached to the cap on an end opposite the grooved blocker portion. The member **624** may be grooved itself to aid in removing the sleeve **602** from the external thermostat block. The cylindrical member **624**, cap **634**, grooved blocker portion **622**, grooves **632** and **633**, groove **629**, cylindrical body **620**, cylindrical cavity **626**, and second cylindrical cavity **628** are all coaxial and share a central longitudinal axis extending through the center of the sleeve **602** lengthwise. The grooved blocker portion **622** and cap **634** together have a length L_2. In a preferred embodiment of the valve **600**, L_2 is 11.557 mm (0.455"). Each groove **632** and **633** have a length L_7 and a diameter D_4. In a preferred embodiment of the valve **600**, L_7 is 1.9558 mm (0.077") and D_4 is 19.3802 mm+/−0.0254 mm (0.763"+/−0.001". Lands **635**, **637**, and **639** of the grooved blocker portion **622** have a diameter D_3. In a preferred embodiment of the valve **600**, D_3 is 21.844 mm (0.860"). Lands **637** and **639** have a length L_6. In a preferred embodiment of the valve **600**, L_6 is 1.524 mm (0.060"). Land **635** has a length L_8. In a preferred embodiment of the valve **600**, L_8 is 3.1242 mm (0.123"). The cap **634** has a length L_3 and a diameter D_1. In a preferred embodiment of the valve **600**, L_3 is 1.4732 mm (0.058") and D_1 is 25.4 mm (1.000"). The member **624** has a length L_4 and a diameter D_2. In a preferred embodiment of the valve **600**, L_4 is 7.62 mm (0.300") and D_2 is 8.255 mm (0.325").

FIG. **16**B shows a rotated view of the sleeve **602** to illustrate the plurality of openings **630** in the groove **629**. A view from the opposite side of the sleeve **602** mirrors the view shown in FIG. **16**B in the preferred embodiment.

Referring now to FIG. **17**, a side view of the piston **614** is provided. The piston **614** has a first cylindrical portion **640** adjacent to and continuous with a second cylindrical portion **642**. A central cylindrical cavity **644** extends through both the first and second cylindrical portions **640** and **642**, and the cylindrical cavity, first cylindrical portion, and second cylindrical portion are coaxial. A free end **646** of the second cylindrical portion **642** may have an outer edge **648** that is beveled. The cavity **644** is open along the free end **646** and an opposing free end **650** along the first cylindrical portion **640**. The piston has a length L_1. In a preferred embodiment of the valve **600**, L_1 is 28.448 mm (1.120"). The first cylindrical portion has a length L_3 and a diameter D_1. In a preferred embodiment of the valve **600**, L_3 is 15.748 mm (0.620") and D_1 is 13.8684 mm (0.546"). The second cylindrical portion has a length L2 and a diameter D3. In a preferred embodiment of the valve **600**, L_2 is 12.7 mm (0.500") and D_3 is 10.8458 mm (0.427"). The cylindrical cavity has length L_1 and a diameter D_2. In a preferred embodiment of the valve **600**, D_2 is 6.9088 mm (0.272"). The beveled outer edge **648** has a length of 0.762 mm (0.030") in a preferred embodiment.

Referring now to FIG. **18**, the thermal bypass valve **600** and constituent components are shown and arranged for insertion into, and operation within, the external thermostat block. O-ring **604** is slipped over the sleeve **602** and secured within the groove **632** and O-ring **606** is slipped over the sleeve and secured within the groove **633**. The bearing ball **610** is inserted within the cylindrical cavity **626** of the cylindrical body **620** of the sleeve **602**. The conical spring **608** is then inserted into the cylindrical cavity **626** such that the smaller diameter end **660** is positioned around a surface of the bearing ball **610**. A larger diameter end **662** of the conical spring **608** is positioned against the free end **650** of the first cylindrical portion **640** of the piston **614**, which is also partially inserted into the cylindrical cavity **626** of the sleeve **602**. The piston O-ring **612** is secured around the second cylindrical portion **642** of the piston **614** adjacent to the first cylindrical portion **640**. The piston **614** and conical spring **608** keep the bearing ball **610** secured within an opening formed at the juncture of the second cylindrical cavity **628** and beveled end **623** of the cylindrical cavity **626** of the sleeve **602**. In a preferred embodiment of the valve **600**, the bearing ball has a diameter of 8.5 mm (0.335").

In operation, fluid traveling through the out-line from the transmission to the cooler either passes through valve **600** via the plurality of openings **630** and second cylindrical cavity **628** and/or around the valve along the groove **629** to continue towards the cooler. This is a default position of the valve **600**, whereby fluid is allowed to flow from the transmission to the cooler and back again to supply cooled fluid to the transmission. The bearing ball **610** blocks access to the first cylindrical cavity **626** in the default position, as the spring **608** holds the bearing ball **610** in contact with an inner circumferential edge of the beveled inner end **623** shared by the second cylindrical cavity **628**. The O-rings **604** and **606** provide sealing edges that keep fluid from escaping through an opening in the thermostatic block through which the valve **600** is installed. When pressure builds up in the out-line, due for example to a blockage in the cooler preventing fluid flow from the cooler to the transmission via the in-line, forces act on the bearing ball **610** to compress spring **608**. This in turn opens a passage from the second cylindrical cavity **628** to the first cylindrical cavity **626** and allows fluid to bypass the cooler. Fluid then flows from the out-line through the plurality of openings **630**, into the second cylindrical opening, around the bearing ball **610**, past the spring **608** through gaps in spring coils, through the cavity **644** of the piston **614** and out of the valve **600** and into the in-line. Once pressure eases in the out-line, the spring **608** expands, forcing the bearing ball **610** to once again seal the connection point between the first and second cylindrical cavities **626** and **628**. The piston **614** slidably engages the cylindrical cavity **626** of the sleeve **626** along at least a partial length of the first cylindrical portion **640** such that fluid cannot escape the sleeve between the sleeve and piston in the cylindrical cavity. The O-ring **612** helps to ensure that fluid does not travel around an outer surface of the piston **614** and must pass through cavities **626**, **628**, and **644** of the valve **600** in order to bypass the cooler. The piston **614** does not move relative to the sleeve **602** during normal operation of the transmission and thermal control system.

Referring now to FIG. **19**, the thermal bypass valve **600** is inserted into the external thermostat block **702** after the OEM thermostatic valve is removed and discarded or recycled. The second cylindrical portion **642** of the piston **614** is inserted into a recessed portion of the external thermostat block **702**, along with the piston O-ring **612**, and the sleeve **602**, two O-rings **604** and **606**, conical spring **608**, and bearing ball **610** are inserted into the external thermostat block such that the valve **600** is contained within the block **702** as shown in FIG. **18**. An OEM snap ring **704** is saved from the OEM thermostatic valve and used to secure the sleeve **602** and remaining components of the valve **600** in place. Together, the thermal bypass valve **600**, OEM snap ring **704**, and external thermostatic block **702** form an improved thermal control system **700**.

Referring to FIG. **25**, an alternative embodiment of a thermal control bypass valve **1200** is shown. This valve **1200** is designed to be inserted into a thermal element located within the cooler itself. The valve **1200** shares similar structures with the valve **600** configured to be inserted into a thermal element located in the cooler lines.

A sleeve **1202** of the valve **1200** has an upper cylindrical portion **1212** and a lower cylindrical portion **1214** defined by a groove **1216** extending there between. The groove **1216** extends around a circumference of the valve **1200** and has a smaller diameter than either the upper cylindrical portion **1212** or the lower cylindrical portion **1214**. An inner cavity **1224** extends within an interior of the valve **1200**, with one of the inner cavity **1224** being closed and another end of the cavity open along a beveled edge **1222** increasing in diameter toward a free end **1220** of the valve **1200**. The beveled edge **1222** is adjacent to and contiguous with the inner cavity **1224** and the free end **1220** of the valve **1200**. At least one opening **1218** in the groove **1216** opens into the inner cavity **1224**, providing fluid access through at least two different openings in the valve **1200**. Two or more openings are preferred in the groove **1216**, to allow fluid access laterally through the valve **1200** by fluid traveling through the out-line, and to allow sufficient amounts of fluid into the inner cavity **1224** and through the free end **1220** during bypass. A first narrow groove **1228** extends around a circumference of the upper cylindrical portion **1212**, and a second narrow groove **1226** extends around a circumference of the lower cylindrical portion **1214**. These grooves **1226** and **1228** are wide enough to securely accommodate and hold O-rings **1208** and **1210**, respectively, which have identical dimensions. A bearing ball **1204** is held partially within the inner cavity **1224** and an area created by a circumference and width of the beveled edge **1222** by a conical spring **1206**. The conical spring **1206** engages the bearing ball **1204** about its surface along a small-diameter end **1240** of the spring. A large-diameter end **1242** of the spring **1206** supports the spring against an inner surface of the thermal element and provides an immovable support upon which the spring compresses during fluid bypass of the cooler.

The beveled end **1222** helps to ensure that the bearing ball **1204** fully closes the opening to the inner cavity **1224** along the free end **1220** of the valve **1200**. As with the valve **600** in previous embodiments, a squared end runs the risk of the bearing ball **1204** getting caught between the spring **1206** and end **1220** without fully blocking fluid access to the inner cavity **1224**. The beveled edge **1222** ensures that, even in the event of a misalignment, the bearing ball **1204** returns to fully sit within the opening at the free end **1220** to prevent and/or restrict fluid flow to the inner cavity **1224**.

A cap **1230** adjacent to the upper cylindrical portion **1212** opposite the groove **1216** provides an outer surface of the valve **1200** after installation and seals the valve within the fluid circuit of the thermal control system by preventing fluid from escaping past the cap. A member **1232** extends from the cap opposite the upper cylindrical portion **1212** to provide a surface for more easily removing and/or installing the valve **1200** in the thermal member.

In operation, the valve **1200** is inserted into a thermal element of the cooler, such that the groove **1216** and at least one opening **1218** are positioned in fluid connection with the out-line of the thermal control system. The cooler thermal element is shaped similar to the thermal block **702** shown in FIG. **19**, but made integral with the cooler instead of positioned in the cooler lines. Fluid traveling through the out-line from the transmission to the cooler either passes through valve **1200** via the at least one opening **1218** and inner cavity **1224** and/or around the valve along the groove **1216** to continue towards the cooler. This is a default, cooler flow position of the valve **1200**, whereby fluid is allowed to flow from the transmission to the cooler and back again to supply cooled fluid to the transmission. The bearing ball **1204** blocks fluid access through the inner cavity **1224** and out the free end **1220** in the default position, as the spring **1206** holds the bearing ball in contact with an inner circumferential edge of the beveled edge **1222** shared by the inner cavity **1224** and free end **1220**. The O-rings **1210** and **1208** positioned within grooves **1228** and **1226** provide sealing edges that keep fluid from escaping along an outer surface of the valve **1200** in either direction along a longitudinal length of the valve (i.e. toward the cap **1230** or free end **1220**). When pressure builds up in the out-line, due for example to a blockage in the cooler preventing fluid flow from the cooler to the transmission via the in-line, forces act on the bearing ball **1204** within the inner cavity **1224** to compress spring **1206**. This in turn opens a passage from the inner cavity **1224**, around the bearing ball **1204**, and through the free end **1220** allowing fluid to flow from the out-line directly into the in-line, bypassing the cooler. Fluid then flows from the out-line through the at least one opening **1218**, into the inner cavity **1224**, around the bearing ball **1204**, past the spring **1206** through gaps in spring coils, and into the in-line. Once pressure eases in the out-line, the spring **1206** expands, forcing the bearing ball **1204** to once again seal the opening at the juncture of the inner cavity **1224** and beveled edge **1222**. Unlike the valve **600**, there is no piston element. The large diameter end **1242** of the spring **1206** rests around an opening of the in-line or similar structure. Even when compressed, fluid is allowed to laterally flow through the spring into the in-line. As thermal elements in the cooler lines and in the cooler are similar, depending on the make and model of a particular transmission, embodiments of valves **600** and **1200** may be interchangeable in structure and may be modified in particular dimensions to fit either thermal element.

As with valves shown in FIGS. **7** and **15**, the valve **600** utilizes a spring wherein the spring is present in the fluid flow of the system and is therefore a differential pressure regulating device instead of a blow-off, or pressure-limit device as provided in FIG. **2**.

The valve **600** is compatible with at least Ford® 6R80/90, 4R70/75, and 6F35 transmissions.

The described valves replace OEM valves to apply embodiments of a process of converting the OEM thermal bypass control system, with the three operating states and features previously described, i.e. fully ON, fully CLOSED, and partial ON, to a two-state system that by default has a state with cooler flow switched ON, and as a secondary state has a safety bypass directly to a cooler return or lube as an emergency state when the cooler and or lines are blocked, pinched, or damaged in such a way as to seriously compromise cooler flow and lube. In the OEM state, when this happens, serious damage results from a starvation of lubricant and adequate cooling. The consequences can range from overheated fluid temperature to severe planetary damage, and possibly vehicle fire if an operator ignores warning lamps and continues to operate the vehicle, especially under heavy load. With the instant embodiments of the process applied to a transmission, lubrication is delivered despite cooler blockage. When the over-temperature warning lamp turns on, the operator may continue to operate the vehicle to return it for repair while averting a complete planetary system crash.

Depending on a particular transmission build, one of two preferred embodiments are employed to improve the thermal control system of an automotive transmission. Both embodiments provide the similar major features, as well as the benefits, that precipitate from them, although the implementation may vary according to vehicle specifics, the location of the thermal control system, and/or type of housing that contains it. One primary feature is full time cooler flow. Cooler fluid fills the cooler and lines immediately on initial fill after transmission rebuild or replacement. Further, there is an accurate fluid level check without a warm-up cycle. Next, there is an emergency safety bypass directly to the lube system to prevent catastrophic planetary failure in the event the cooler is restricted or blocked, or lines are pinched or smashed. There are also lower average operating temperatures, typically between 62.778-73.889° C. (145-165° F.), which is about a 21.111° C. (70° F.) reduction from OEM thermal control temperatures.

The first embodiment of the method includes providing a flow-blocking device that functions as a blow-off or pressure limit valve. FIG. **24** illustrates a thermal control system in a transmission pump implementing a process with a pressure limit valve. The OEM valve is removed and replaced with a valve like the one shown in FIGS. **2-5** of the instant disclosure. The valve is installed into the pump replacing the OEM valve. In a default state, or position at vehicle start-up, the relief valve **1106** blocks a direct path **1134** between in-line **1120** and out-line **1118** through the thermal element. This forces cooler fluid to flow from other valves **1128** in the pump **1132**, through out-lines **1134** and **1124**, to the cooler **1122**, through in-lines **1126**, **1120**, **1121**, and **1130** and back through gears **1116**, pump **1132** and other valves **1128**, If PSI within the fluid circuit of in-lines and out-lines reaches the blow-off PSI, the relief valve **1106** is forced outward away from a center of the pump **1132** and the spring **1108** is compressed, thus allowing for cooler flow from the valves **1128**, through out-line **1134**, line **1118**, line **1112**, the direct path **134**, line **1121**, through the gears **1116**, in-lines **1130**, and back into the valves **1128** to include the thermal element in the circuit and bypass the cooler **1122**. This scenario would occur if a blockage in the cooler **1122** prevents fluid flow through the cooler. When pressure decreases in the out-lines, forces on the relief valve **1134** decrease and the spring **1108** expands to push the relief valve back towards the blocker valve **1104**. This re-establishes flow from the pump and transmission to the cooler. The spring **1108** is in a vented area between the relief valve **1134** and cap **1110**, so there is no counterbalance oil pressure on the spring side of the relief valve. Therefore, the blow-off PSI is equal to the spring tension divided by the area of the end of the valve **1134**. The area of the valve and spring tension can then be altered to achieve a preferred blow-off PSI at which point the valve will switch the system between the cooler bypass configuration or the configuration allowing flow to the cooler.

A second embodiment of the method includes replacing an OEM valve in the thermal element with an intermediary or differential pressure regulating device. This pressure provides a counter-balance force on equal areas. FIGS. **20**A and **21**A show an OEM valve in a default and altered position, respectively. FIGS. **20**B and **21**B show the OEM valve installed in a thermal block in either the default or altered position, respectively, to illustrate a simplified thermal control fluid circuit. At start-up in an OEM system **900**, the valve **800** is retracted per FIG. **20**A, either fully or partially depending on fluid temperature as previously discussed, allowing cooler fluid flow from transmission **912** through the out-line **904**. The fluid then flows through an opening **914** into an inner cavity **908** of the thermal element **910** and back to the transmission **912** via an in-line **906**. This state of the OEM thermal control system **900** is shown in FIG. **20**B and ensures that a majority of the fluid through the out-line **904** bypasses the cooler **902**. The design of the OEM valve **800** and system **900** assumes that full flow through the cooler would only be necessary when fluid temperature becomes too high. In such a case, silicon in the valve **800** expands, causing the valve to lengthen and block the entrance to the inner cavity **908** of the thermal element **910**. The lengthened OEM valve **800** is shown in FIG. **21**A. As seen in FIG. **21**B, the altered OEM valve **800** blocks liquid flow through the opening **914**. Cooler fluid would then be forced from the transmission **912** through the out-line **904** to the cooler **902** and transported back to the transmission via the in-line **906**.

The second embodiment of the method pertains to environments where there is fluid pressure, or cooler return, in the area that contains the spring, as opposed to the first embodiment where the spring is in vented area separate from the fluid circuit. In this case the formula now includes cooler return PSI, so is expressed as follows:

PRESSURE 1 (to cooler)×AREA=SPRING+PRESSURE 2 (cooler return).

This formula is used to determine corresponding dimensions and structures of the valves of this embodiment to set a preferred pressure differential within the system for the system to maintain via the valve.

FIGS. **22** and **23** illustrate the application of the second embodiment of an inventive system **1000** improving the OEM system **900** shown in FIGS. **20**B and **21**B. A transmission **1012**, out-line **1004**, in-line **1006**, cooler **1002**, and thermal element **1010**, including a valve **1020**, are shown. A default state, or cooler flow ON, of the instant method is shown in FIG. **22**. An entrance **1014** to a bypass chamber **1008** of the thermal element **1010** from the out-line **1004** is closed by the valve **1020**, forcing fluid flow from the transmission **1012** directly to the cooler **1002** and back to the transmission via the in-line **1006** at start-up of the vehicle. This ensures a proper supply of cooled liquid lubricant during start-up. When cooler flow through the cooler **1002** is blocked, a pressure differential is created such that pressure in the in-line **1006** approaches zero. In that instance, the system **1000** behaves as shown in FIG. **23** such that the valve **1020** is compressed by the fluid pressure in the out-line **1004**. This allows fluid through opening **1014** and into the bypass chamber **1008**, allowing fluid to flow from the out-line **1004**, through the bypass chamber and back to the transmission **912** via the in-line, bypassing the cooler **1002**.

For example, assuming a particular transmission with all OEM parts regulates a minimum main line pressure of 448.159 kPa (65 PSI), and characteristically delivers 2.0 GMP at 275.79 kPa (40 PSI) from the converter to the cooler lines (cooler out-line), the cooler in-line back to the transmission delivers the same 2.0 GPM at 137.895 kPa (20 PSI). If the cooler was to become blocked, then the cooler out-line would rise to nearly equal to main line PSI, so from 275.79 kPa (40 PSI) to 448.159 kPa (65 PSI), and the cooler in-line would drop toward zero.

Conversely, a valve of the instant invention installed with a spring calibrated for a preferred pressure differential of 275.79-310.264 kPa (40-45 PSI) would allow for a bypass supply of 1.0-2.0 GPM fluid at 103.421-172.369 kPa (15-25 PSI), at 413.685-448.159 kPa (60-65 PSI) in the cooler OUT line, through the bypass chamber and back to the transmission through the cooler IN line. This keeps a steady supply of fluid into the planetary system. Maintaining a pressure differential range of 241.317-379.212 kPa (35-55 PSI) is desired in this embodiment of the process, and a pressure differential of 310.264 kPa (45 PSI) is most desired. In this situation, the fluid through the bypass chamber will be hotter than in a default state, but crucially an adequate supply of lubrication will continue to be provided to prevent catastrophic planetary failure. Valves of FIGS. **7** and **15**, and their corresponding embodiments may be employed in the instantly described method for improving fluid flow in the thermal control system. Such valves include fluid pressure, or cooler return, in the area that contains the spring.

Importantly, the embodiments of the present invention operate on fluid pressure in the cooler lines, either or both of the out-lines and in-lines. High pressure or pressure differentials can be obtained in either extreme heat or in extreme cold. Fluid in the lines of the system approaching either transitional state, between liquid-solid or liquid-gas states, can expand within the lines of the system and increase pressure necessary to instigate a bypass of the cooler. Most discussion so far has assumed high operating temperatures causing increased pressure, or non-temperature related pressurizing events, such as blockages in the cooler. However, extreme cold causes fluid in the lines to begin to gel into a viscous liquid and toward a solid. This slows or prevents cooler flow in the cooler and/or lines. Therefore, systems of the instant invention allow bypass of the cooler under extreme cold conditions and revert back to allowing flow through the cooler under normal operating temperatures, typically once the fluid melts from a gel form.

FIGS. **26**A-**26**E provide views of a ZF 8 transmission **1320**, a corresponding OEM thermal bypass device **1300**, and corresponding engine coolant warmer **1340**. The ZF 8 transmission **1320** and OEM thermal bypass device **1300** were originally designed to ideally maintain a partial mix, or between full cooler flow and full cooler bypass. At engine start, a silicon pack **1304** of the OEM thermal bypass device **1300** is in a contracted position and cooler bypass through the engine coolant warmer **1340** is fully open. Due to flow through the cooler lines acting as a resistor to cooler fluid flow, the transmission fluid travels through opening **1330**, into the warmer via hole **1342** where hot engine coolant entering and leaving the warmer via lines **1348** and **1350** is used to heat the transmission fluid, which then passes through hole **1344** of the warmer and back into the transmission **1320** via hole **1332**. As the transmission fluid is heated in this manner, the silicon in the silicon pack **1304** is likewise heated and begins to expand. The silicon expanding causes an extender pin **1302** to extend and push a seat **1306** to close against opening **1328** inside the transmission **1320**, the opening **1328** leading to opening **1330**. This seals off transmission fluid flow to the warmer **1340**, thereby forcing fluid flow to the cooler via opening **1327** and cooler flow opening **1326** and cooler flow return opening **1324**. As the silicon in the silicon pack **1304** cools, the seat **1306** unseals slightly to ideally allow some fluid flow to the warmer **1340** via opening **1328**, **1330** and some fluid flow to the cooler via opening **1327** to maintain a constant preset temperature.

However, this configuration is only ideal if the corresponding vehicle is at a steady cruising speed in top gear with the converter clutch applied. Otherwise, transmission temperature constantly fluctuates as workload and driving conditions change. The silicon pack **1304** is constantly expanding and contracting which over time wears out the silicon. Time, contaminants in the fluid, and inherent fragility leads to pin **1302** sticking, the seal **1306** wearing out, and the silicon pack **1304** rupturing. This in turn causes transmission **1320** overheating, burnt fluid, and over-expansion of valves, solenoids, and other components, which all cause malfunction or engine failure.

The thermal bypass valve **1400** of FIG. **27** has been designed to overcome the issues present in the design of the OEM thermal bypass device **1300**. Unlike the other thermal bypass valve embodiments described in this disclosure, the thermal bypass valve **1400** corresponds to a transmission that regulates cooler fluid feed at a fixed value, preferably, but not limited to, approximately 172.37 kPa (25 PSI), which includes for example the ZF 8 transmission **1320**. This means that the cooler regulator valve in the main control valve body of the transmission regulates and delivers cooler flow at 172.37 kPa regardless of the flow rate of transmission fluid. Consequently, the design of the thermal bypass valve **1400** cannot rely on changes in internal pressure, i.e. changes in transmission fluid pressure, to operate bypass functions of the thermal bypass valve unlike other thermal bypass valves discussed herein. The thermal bypass valve **1400** therefore includes at least three manual, user-selectable positions of the thermal bypass valve to correspondingly select desired bypass states. Such a configuration of the thermal bypass valve **1400** is applicable to other transmissions designed to provide a fixed transmission fluid pressure regardless of fluid flow rates.

The thermal bypass valve **1400** includes a retainer clip **1402**, a plug **1404**, one or more plug O-rings **1405**, a plunger **1406**, one or more plunger O-rings **1407**, a flange **1409**, a first spring **1408**, a blocker valve **1410**, and a second spring **1412**. The thermal bypass valve **1400** is generally cylindrical, with each separate component having same or different diameters relative to each other. Further, components **1402**, **1404**, **1406**, **1410**, and **1412** may have their own sub-elements with different diameters. The exact shape and

diameter of the thermal bypass valve **1400** may vary to conform to the shape and size of a corresponding bore of the transmission to direct and control fluid flow as described herein. In operation, the thermal bypass valve **1400** is oriented as shown in FIG. **28**, with the plunger inserted through and held within an opening **1419** of a cylindrical upper member **1414** of the plug **1404** along end **1424***a*, except that the clip **1402** may by further attached to the plug **1404** and the plunger **1406**, The opening **1419**, together with an opening **1415** along opposite end **1424***b*, forms an inner cavity **1417** of the plug **1404**. The plunger **1406** extends through the inner cavity **1417** of the plug **1404** such that opposite ends **1436***a*, **1436***b* of the plunger extend beyond either end **1424***a*, **1424***b* of the plug. The one or more plug O-rings **1405** are each secured in a corresponding groove **1422** located along an outer circumference of a lower member **1418** of the plug **1404**. The one or more plunger O-rings **1407** are each secured in a corresponding groove **1430** located along an outer circumference of the plunger **1406**. The flange **1409** is secured within a flange groove **1432** located along the outer circumference of the plunger **1406**, the flange groove **1432** positioned between the one or more grooves **1430** and the end **1436***b*.

The first spring **1408** is positioned between and against the plunger **1406** and the blocker valve **1410**. In the preferred embodiment of the thermal bypass valve **1400**, the first spring **1408** is positioned against the flange **1409** secured to the plunger **1406** and a lip **1443** along an outer surface of the blocker valve **1410**. The second spring **1412** is positioned against and between the blocker valve **1410** and an inner surface **1329** of the transmission bypass valve bore beyond opening **1328**. The retainer clip **1402** is optionally secured within a retainer clip groove **1420** on an outer circumference of the upper member **1414** of the plug **1404** and within one of several retainer clip grooves **1428** along the outer circumference of the plunger **1406**. The retainer clip **1402** may optionally not be attached to the plug **1404** and plunger **1406** in operation, namely when the thermal bypass valve **1400** is in a full bypass configuration. The retainer clip **1402** therefore preferably has at least two areas of attachment which are secured together to prevent movement of the two areas of attachment relative to each other when secured to plug **1404** and plunger **1406**.

When installing the thermal bypass valve **1400**, the OEM thermal bypass device **1300** is first removed from the bypass valve opening **1322** located on the transmission **1320**, as shown in FIG. **28**. The second spring **1412** is then inserted through the opening **1322** and through opening **1328**, which leads to the opening **1330**, and against inner surface **1329**, Next, the blocker valve **1410** is inserted along end **1446***a*, with end **1446***b* directed towards the opening **1322**. A blocker portion **1440** of the blocker valve **1410** is inserted coaxially through and flush with the opening **1328**, causing the second spring **1412** to contract. The first spring **1408** is then inserted such that one end **1408***b* of the first spring is slid over a receiving portion **1442** of the blocker valve **1410** and against the lip **1443**, such that the first spring is preferably coaxial with the blocker valve and second spring **1412**.

Alternatively, the first spring **1408** may be inserted such that one end **1408***b* of the first spring is slid over a receiving portion **1442** of the blocker valve **1410** before either the first spring or blocker valve are inserted into the transmission **1320**, with both then being inserted together at the same time.

The plunger **1406**, with the plunger O-rings **1407** and the flange **1409** secured within the grooves **1430** and **1432**, respectively, is then inserted through the opening **1322**. The end **1436***b* of the plunger **1406** is inserted first with a partial length of the plunger equal to the length between the flange **1409** and the end **1436***b* inserted within an opposite end **1408***a* of the first spring **1408**, such that the first spring **1408** and plunger are preferably coaxial. The plug **1404**, with the plug O-rings inserted within grooves **1422** is then inserted into the transmission **1320** along the end **1424***b* such that the lower member **1418** is inserted within the opening **1322**. Simultaneously, the plug **1404** is secured around the plunger **1406** by inserting the end **1436***a* of the plunger through the opening **1415**, into the inner cavity **1417**, and through the opening **1419** such that the end **1436***a* of the plunger extends beyond the end **1424***a* of the plug. If applicable, the retainer clip **1402** is then secured to the plug **1404** and plunger **1406** at the desired location depending upon the user-selected bypass configuration.

Alternatively, the plug **1404** and plunger **1406**, and retainer clip if applicable, may be secured together in the same fashion before either of the plug or plunger are inserted through the opening **1322**, In this manner, the plug **1404** and plunger **1406** can be inserted together as one unit.

An OEM retainer clip or similar known mechanism for securing the thermal bypass valve within the valve bore, not shown, may be used to secure the thermal bypass valve **1400** within the valve bore and transmission **1320**, by securing within the opening **1322** and against a cap **1416** of the plug **1404**.

FIG. **29** provides a diagram of three different bypass configurations BP_1, BP_2, and BP_3. However, the structure of the thermal bypass valve **1400** may allow for any number of different bypass configurations between full cooler flow and full bypass. For each of the three bypass configurations BP_1, BP_2, and BP_3 shown by way of example in FIG. **29**, the position of the plunger **1406** relative to the plug **1404** affects the relative position of the blocker valve **1410** relative to the opening **1330** into the engine coolant warmer **1340**. In the bypass configuration BP_1, the thermal bypass valve **1400** is configured to block all transmission fluid flow to the engine coolant warmer **1340** and force all transmission fluid flow to the cooler. In this configuration, the plunger **1406** is fixed deeper in the transmission thermal bypass valve bore by positioning the retainer clip **1402** in groove **1428***a* and groove **1420**. The flange **1409** is therefore fixed closer to the lip **1443** which forces the first spring **1408** to contract. However, as the plunger **1406** is fixed in position, but the blocker valve **1410** is not fixed relative to the plunger or plug **1404**, the first spring **1408** acts on and moves the blocker valve **1410** to completely cover the opening **1330** with the blocker portion **1440** to prevent fluid flow through the engine coolant warmer **1340**. The counteracting forces of the springs **1408** and **1412** hold the blocker valve **1410** in place between them over the opening **1330**. In other words, the thermal bypass valve is configured for full cooler flow, which provides maximum transmission fluid flow and cooling.

In bypass configuration BP_2, the thermal bypass valve **1400** is configured to allow partial transmission fluid flow to the engine coolant warmer **1340** and partial fluid flow to the cooler. This is also referred to as mixed flow or partial mix fluid flow. The retainer clip **1402** is secured within groove **1428***b* and groove **1420**, which secures the plunger **1406** and the flange **1409** farther away from the blocker valve **1410** and lip **1443** relative to the positioning of the same elements in bypass configuration BP_1. As there is less force acting on the first spring **1408**, the blocker valve **1410** is moved such that the blocker portion **1440** only partially covers the opening **1330**. The counteracting forces of the springs **1408** and **1412** hold the blocker valve **1410** in place between them to partially cover the opening **1330**. This configuration allows limited fluid flow through the engine coolant warmer **1340** while also allowing fluid flow through the cooler. This will result in elevated operating temperatures relative to the bypass configuration BP_1.

In bypass configurations BP_3, thermal bypass valve **1400** is configured to allow full fluid flow to the engine coolant warmer **1340**, or full cooler bypass. The retainer clip **1402** is not attached to the plug **1404** and plunger **1406**, which allows the second spring **1412** to push the blocker valve **1410** away from blocking the opening **1330**. The first spring **1408** then pushes the plunger **1406** back toward the opening **1322**. The plunger O-rings **1405** seal off fluid from escaping between the plug **1404** and plunger **1406**, and the flange **1409** prevents the plunger **1406** from being expelled from the bore and plug once it contacts a back surface of the inner cavity **1417**. The thermal bypass valve **1400** is in full bypass in this configuration. The opening **1327** to the cooler is not blocked, but as the cooler line acts as a fluid circuit resistor, the transmission fluid will take the path of least resistance through the engine coolant warmer **1340**. This configuration is useful to preheat the transmission **1320** to above 73.888° C. (165° F.) to meet a vehicle control module's temperature requirement to initialize a computer RE-LEARN state after a transmission repair or rebuild.

FIGS. **30**A and **30**B show a preferred embodiment of the plug **1404** in more detail. The plug **1404** extends longitudinally along axis C-C. The plug **1404** includes the upper member **1414** and lower member **1418** on either side of a cap **1416**, the upper member and lower member extending in opposite directions away from the cap, which is wider in diameter than both the upper member and lower member. The cap **1416**, upper member **1414**, and lower member **1418** are cylindrical, with the lower member **1418** shaped to fit the bore of the transmission thermal valve and the cap shaped to conform to opening **1322**. However, the upper member **1414** can vary in shape and design. The upper portion **1418** has the groove **1420** extending continuously along its outer circumference. The groove **1420** is configured to hold and secure the retainer clip **1402** to the upper portion **1414** when needed. The lower member **1418** preferably has two grooves **1422**, each configured to hold a corresponding plug O-ring **1405**, The plug O-rings **1405**, once properly installed in a corresponding groove **1422**, provide a co-efficient of friction against the bore surface to help secure the plug **1404** in the transmission bore. The plug O-rings **1405** also help prevent fluid leakage between the plug **1404** and bore. The plug O-rings **1405** are preferably elastic and slightly smaller in diameter than the corresponding grooves **1422** to ensure a tight fit and are made from rubber or rubber-like materials or any material with similar and qualifying characteristics, such as exhibiting both a viscous and elastic nature under deformation. The opening **1419** includes a cylindrical channel extending through the upper member **1414** and adjacent to opening **1415**, which includes another channel extending through the lower member **1418**. Together, the openings **1415** and **1419** form the inner cavity **1417** extending along the length of the plug **1404**, A bevel **1413** is preferably placed along the opening **1419** where that opening meets and is contiguous with opening **1415** in the inner cavity **1417**. This bevel **1413** is configured to allow a plunger O-ring **1407** to elastically deform against the bevel such that the O-ring is not sheared, damaged, improperly moved or otherwise rendered inoperable in preventing fluid flow between the plug **1404** and plunger **1406**.

The plug **1404** has preferred dimensions. The overall length L_1 of the plug **1404** is 21.717 mm (0.855"). The upper member **1414** has a diameter D_2 of 15.875 mm (0.625") and a length L_2 of 10.160 mm (0.400"). The groove **1420** has a diameter D_3 of 13.081 mm+/−0.0254 mm (0.515"+/−0.001") and a length of L_8 of 1.016 mm (0.040"). A length L_9 of the upper member **1414** above the groove **1420** is 2.540 mm (0.100"). The cap **1416** has a diameter D_1 of 25.146 mm (0.990") and a length L_4 of 1.4732 mm (0.058"). The lower member **1418** has a diameter D_4 of 21.844 mm (0.860") and an overall length L_5 of 10.1092 mm (0.398"). The grooves **1422** each have a diameter D_5 of 19.3802 mm+/−0.0254 mm (0.763"+/−0.001") and a length L_7 of 1.9558 mm (0.077"). The portion of the lower member **1418** between grooves **1422** and along the end **1424***b* each have a length L_6 of 1.524 mm (0.060"). The opening **1419** has a diameter D_6 of 11.1252 mm (0.438"). The opening **1415** has a diameter D_7 that may vary but is otherwise at least as large as diameter D_6 and at least smaller than diameter D_4. The bevel **1413** extends from a diameter of 11.0998 mm (0.437") to 11.684 mm (0.460").

FIG. **30**C shows the plunger **1406** from the side extending along a longitudinal axis A-A. The plunger **1406** is cylindrically shaped and longitudinally extending along axis A-A to slidably engage the plug **1404** through the opening **1419**. In the preferred embodiment, the plunger **1406** has retaining clip grooves **1428** near end **1436***a*, including groove **1428***a* corresponding to full cooler flow and groove **1428***b* corresponding to mixed flow. Additionally grooves **1428** may be present if more than three bypass configurations are desired. The groove **1432** is located along the plunger **1406** near end **1436***b* and is configured to removably secure the flange **1409**. Alternatively, the flange **1409** may be permanently attached about the plunger or be an extension thereof. Between grooves **1428** and groove **1432** are one or more grooves **1430** configured to secure a corresponding plunger O-ring **1407** about the plunger **1406** to prevent the plunger from moving longitudinally through the plug **1404** along an entire length of the plunger once secured within the opening **1419**. Along end **1436***b*, the plunger **1406** has a tapered surface **1434** for interacting with the first spring **1408**. The plunger **1406** may alternatively not include the tapered surface **1434**.

The plunger **1406** has an overall length L_{10} of 48.387 mm (1.905"), with section **1406***a* having a length L_{13} of 2.540 mm (0.100"), section **1406***b* having a length L_{14} of 4.2164 mm (0.166"), section **1406***c* having a length L_{15} of 8.6106 mm (0.339"), section **1406***d* having a length L_{16} of 1.27 mm (0.050"), section **1406***e* having a length L_{17} of 15.24 mm (0.600"), section **1406***f* having a length L_{18} of 2.032 mm (0.080"), and section **1406***g* having a length L_{19} of 7.620 mm (0.300"). The plunger **1406**, including sections **1406***a*, **1406***b*, **1406***c*, **1406***d*, **1406***e*, and **1406***f*, has a diameter D_9 of 11.0744 mm (0.436"). The retainer clip grooves **1428** have a diameter D_{10} of 5.9944 mm (0.236"). The plunger O-ring grooves **1430** each have a diameter D_{11} of 7.9248 mm+/−0.0254 mm (0.312"+/−0.001"). The flange groove has a diameter D_{12} of 7.366 mm (0.290"). The end **1436**B of the plunger **1406** has a diameter D_{13} of 5.080 mm (0.200"). A length L_{12} of the plunger **1406** between, and including, the grooves **1430** and the groove **1432** is 20.320 mm (0.800"). A length L_{11} of the plunger **1406** between the end **1436**A and groove **1432** is 37.719 mm (1.485").

FIG. **30**D shows the blocker valve **1410** of the preferred embodiment of the thermal bypass valve **1400** extending longitudinally along axis B-B. The blocker valve includes the cylindrical blocker portion **1440** and the cylindrical receiving portion **1442** secured together about respective ends such that the two portions are preferably coaxial. A lip **1443** is formed where the two portions **1440** and **1442** meet, which provides a surface for the first spring **1408** to act against, with the other surface being the flange **1409**. The lip is preferably a perpendicular surface relative to the longitudinal length of the block valve **1410** along axis B-B. An inner channel **1444** extends longitudinally through the blocker valve **1410** with an opening at either end **1446***a* and **1446***b*. The inner channel **1444** allows fluid flow past the blocker valve **1410** and to the engine coolant warmer **1340** when in a full bypass or mixed flow configuration. The blocker valve has an overall length L_{23} of 28.448 mm (1.120"). The receiving portion **1442** has a length L_{24} of 12.700 mm (0.500") and a diameter D_{15} of 10.8458 mm (0.427"). The blocker portion has a length L_{25} of 15.748 mm (0.620") and a diameter D_{14} of 13.8684 mm (0.546"). The inner cavity **1444** has a diameter D_{16} of 6.9088 mm (0.272").

FIG. **31** shows an alternate embodiment of the thermal bypass valve **1400**, which includes the depicted plunger **1500**. In this embodiment a retaining clip **1402** is not used. As such, retaining clip grooves **1428** are not needed or present on the plunger **1500**. Instead, a screw **1514**, or similar threaded member, is present in the plug **1404** and threadingly engages a threaded surface **1504** of the plunger, which replaces the grooves **1428**. Rotating the screw **1514** in turn causes the plunger **1500** to extend or retract relative to the plug **1404**. The screw **1514** can be adapted to receive known tools, such as Allen wrenches and screwdrivers, including Phillips, slotted, Robertson, and torix screwdrivers. The plunger **1500** otherwise retains the same structure as plunger **1400**, including O-ring grooves **1506**, a flange groove **1508**, a tapered surface **1510**, and ends **1512***a* and **1512***b*. All other components of the thermal bypass valve **1400** remain the same as previously discussed.

Another embodiment of the thermal bypass valve **1400** is shown in FIGS. **32**A-**32**B. In this embodiment, the plunger **1600** does not have the retaining clip grooves **1428** as the retaining clip **1402** is not used. Instead, the plunger **1600** is moved relative to the plug **1620** via a screw **1614** or other fastener inserted through a hole **1630** in the upper member **1622** of the plug **1620**. The screw **1614** inserts through the hole **1630** and into a longitudinal groove **1616** in a circumferentially grooved section **1606**, which contains threading or other textured surface configured to interact with the screw. Rotation of the screw **1614** while in the groove **1616** cause forward or backward longitudinal movement of the plunger **1600** relative to the plug **1620**. The plunger **1600** otherwise retains the same structure as plunger **1406**, including O-ring grooves **1608**, a flange groove **1610**, a tapered surface **1604**, and ends **1612***a* and **1612***b*. The plug **1620** otherwise retains the same structure as plug **1404**. All other components of the thermal bypass valve **1400** remain the same as previously discussed.

Yet another embodiment of the thermal bypass valve **1400** includes plunger **1700**, with a threaded portion **1710** beginning along end **1712***a*. A nut, not shown, is secured along the threaded section **1710** and against the plug **1404** to secure the plunger at the desired position, instead of using the retaining clip **1402** and grooves **1428** or a fastener, as in FIGS. **31**, **32**A, and **32**B. The plunger **1600** otherwise retains the same structure as plunger **1406**, including O-ring grooves **1704**, a flange groove **1706**, a tapered surface **1708**, and ends **1712***a* and **1712***b*. All other components of the thermal bypass valve **1400** remain the same as previously discussed.

In other embodiments of the thermal bypass valve, the flange **1409** may be integrally secured about and extending from the plunger **1406**, instead of fitting into the flange groove **1432**. In such an embodiment, the flange groove **1432** would be unnecessary.

Further, springs **1408** and **1412** may be of different lengths and have different spring forces to achieve the desired configurations of full cooler flow, full bypass, and different stages of mixed cooling and bypass. Likewise, the exact position of the plunger **1406** in the thermal valve bore can vary when configured to achieve full cooler flow, full bypass, or mixed flow depending on spring lengths and spring forces. All these factors may vary to achieve the proper positioning of the blocker valve **1410**. It is the positioning of the blocker valve **1410** which ultimately determines full cooler flow, full bypass, or mixed flow in the transmission **1320** by fully blocking opening **1330**, not blocking opening **1330** at all, or partially blocking opening **1330**, respectively.

Certain additional elements that may be needed for operation of some embodiments have not been described or illustrated as they are assumed to be within the purview of those of ordinary skill in the art. Moreover, certain embodiments may be free of, may lack and/or may function without any element that is not specifically disclosed herein.

Features of embodiments and discussed herein may be combine with features of other embodiments discussed herein in some examples of implementation. Although various embodiments and examples have been presented, this disclosure describes, but does not limit, the invention. Various modifications and enhancements will become apparent to those of ordinary skill in the art and are within the scope of the invention.

We claim:

1. A thermal bypass valve, comprising:

a plug having an inner cavity longitudinally extending between two oppositely oriented openings and a cap configured to prevent fluid flow between the thermal bypass valve and a valve bore in a transmission, the plug configured to insert into an opening of the valve bore;

a longitudinally extending plunger configured to be secured coaxially through the inner cavity of the plug and extending beyond the two oppositely oriented openings of the plug;

a blocker valve having and a fluid channel longitudinally extending between two oppositely oriented blocker valve openings;

a first spring configured to be secured between the plunger and the blocker valve; and

a second spring configured to be secured between the blocker valve and a surface of the valve bore,

wherein, when the thermal bypass valve is installed in a transmission, a position of the plunger relative to the plug is manually changeable to enable full transmission fluid flow to a fluid cooler, full transmission fluid bypass of the fluid cooler, and mixed transmission fluid flow to the fluid cooler and to a fluid warmer.

2. The thermal bypass valve of claim **1**, further comprising a flange secured to the plunger adjacent to a first end of the plunger, the flange configured to engage the first spring and extending perpendicularly from the plunger relative to a longitudinal axis of the plunger.

3. The thermal bypass valve of claim **2**, wherein the blocker valve has a lip for engaging the first spring.

4. The thermal bypass valve of claim **1**, wherein the plug further comprises an upper member and a lower member, the upper member and lower member longitudinally extending on either side of the cap.

5. The thermal bypass valve of claim **4**, wherein the lower member has one or more grooves extending circumferentially along an outer surface of the lower member.

6. The thermal bypass valve of claim **5**, further comprising one or more plug O-rings, each plug O-ring of the one or more plug O-rings corresponding to a groove of the one or more grooves extending along the outer surface of the lower member.

7. The thermal bypass valve of claim **1**, wherein the plunger has one or more grooves extending circumferentially along an outer surface of the plunger.

8. The thermal bypass valve of claim **7**, further comprising one or more plunger O-rings, one plunger O-ring of the one or more plunger O-rings corresponding to a single groove of the one or more grooves extending circumferentially along the outer surface of the plunger.

9. The thermal bypass valve of claim **1**, wherein the plunger has a tapered surface along the first end.

10. The thermal bypass valve of claim **1**, further comprising a retaining clip,

wherein the plug has a retaining clip groove extending circumferentially along an outer surface of an upper member,

wherein the plunger has a plurality of flow grooves extending circumferentially along an outer surface of the plunger, and

wherein the retaining clip is configured to simultaneously secure in the retaining groove and one flow groove of the plurality of flow grooves.

11. The thermal bypass valve of claim **1**, further comprising a means to manually change a position of the plunger relative to the plug when the plunger is secured coaxially through the inner cavity of the plug and extending beyond the two oppositely oriented openings of the plug.

12. A method of manually regulating transmission cooler flow, comprising:

installing the thermal bypass valve of claim **1** in the transmission; and

manually moving the plunger relative to the plug to set various bypass configurations, which changes a position of the blocker valve between the first spring and second spring relative to an opening to a fluid warmer,

wherein changing the position of the blocker valve the opening to the coolant warmer allows for the full transmission fluid flow to the fluid cooler, full transmission fluid bypass of the fluid cooler, and mixed transmission fluid flow to the fluid cooler and to a fluid warmer.

13. The method of claim **12**, further comprising removing the original equipment manufacturer (OEM) thermal bypass valve from the transmission before installing the thermal bypass valve.

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